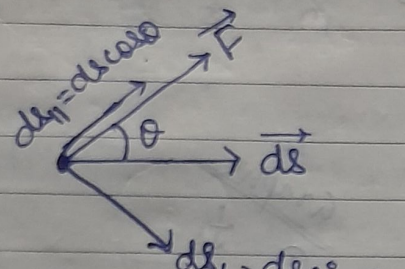
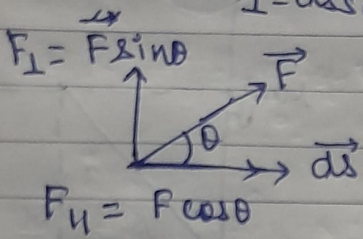
Work, Power and Energy

Work: A force will do some work if there is <u>displacement of point of application of force</u> in any direction other than perpendicular to force.

requirement of Work - ① force ② displacement of POAF in dirn other than ⊥ to force.



$$dW = \vec{F}\cdot\vec{ds}$$
$$= F\,ds\cos\theta$$
$$= F\,(ds\cos\theta) = (F\cos\theta)\,ds$$
$$= F\,ds_{(\text{parallel to } F)}$$
$$= F_{II}\,ds$$



→ Work done is scalar

→ Can be +ve ($\theta < 90°$), -ve ($\theta > 90°$), zero ($\theta = 90°$)

→ Unit ⇒ Nm = Joule (J)

CGS = dyne-cm = ergs

1 Joule = 1 Nm = 10^5 dyne · 10^2 cm = 10^7 ergs

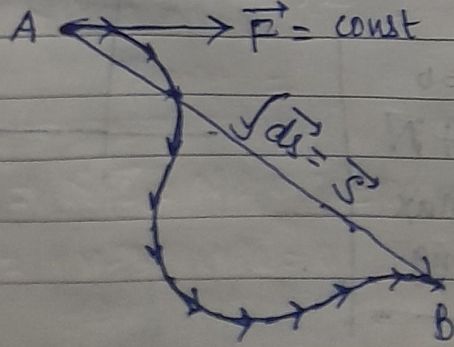
$\therefore$ $\boxed{1\,J = 10^7 \text{ ergs}}$

→ Dimensions: $[ML^2T^{-2}]$

→ Work is reference frame dependent.

$$\boxed{W = \int dW = \int \vec{F}\cdot\vec{ds}}$$

Work Done by Constant Force:



$$W = \int dW = \int \vec{F}\cdot\vec{ds} = \vec{F}\cdot\int\vec{ds}$$
$$= \vec{F}\cdot\vec{s}$$

$$\boxed{W = Fs\cos\theta}$$

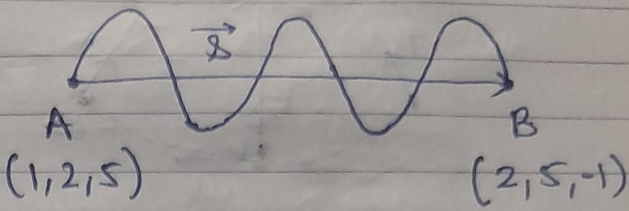
Work done by constant force is:

→ Independent of path

→ In a closed path W.d. by constant force is zero (as disp. = 0)

Ex $\vec{F} = (2\hat{i} + 3\hat{j} - 4\hat{k})$ N

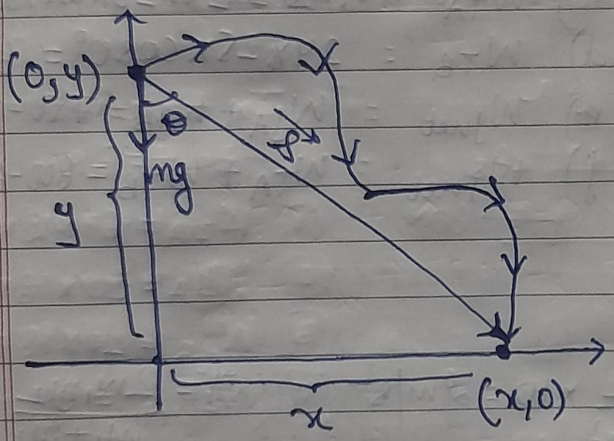
Find W.D. by $\vec{F}$.



$\vec{s} = (\hat{i} + 3\hat{j} - 6\hat{k})$ m

$W = \vec{F} \cdot \vec{s} = (2\hat{i} + 3\hat{j} - 4\hat{k}) \cdot (\hat{i} + 3\hat{j} - 6\hat{k}) = 2 + 9 + 24 = 35$ J

Work Done by Gravity:



$W = \vec{F} \cdot \vec{s}$

$= -mg\hat{j} \cdot (x\hat{i} - y\hat{j})$

$\boxed{W_g = mgy}$

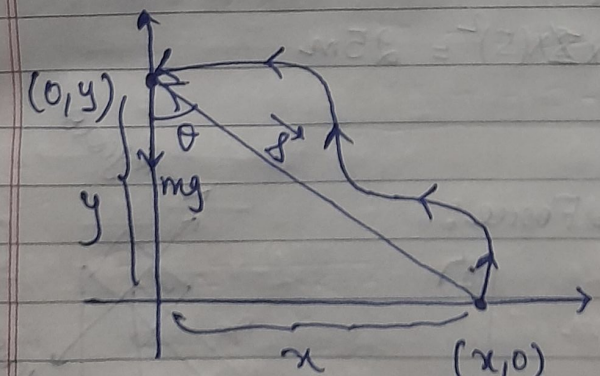
$W_g = mgy$ = W.d. by gravity

$W = fs\cos\theta$

$= (mg)(\sqrt{x^2+y^2})\left(\frac{y}{\sqrt{x^2+y^2}}\right)$

$W = fs_\perp$

$= (mg)y$



$W = \vec{F} \cdot \vec{s}$

$= -mg\hat{j} \cdot (-x\hat{i} - y\hat{j})$

$\boxed{W_g = -mgy}$

∴ If particle comes down: W_g = +ve

If particle goes up ⇒ W_g = -ve

Work Done by Tension:

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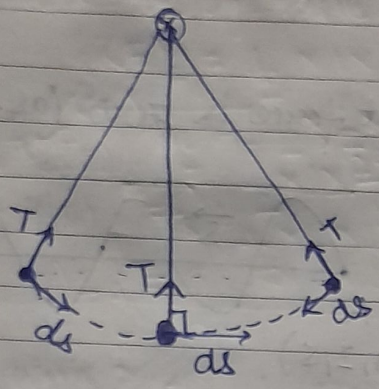
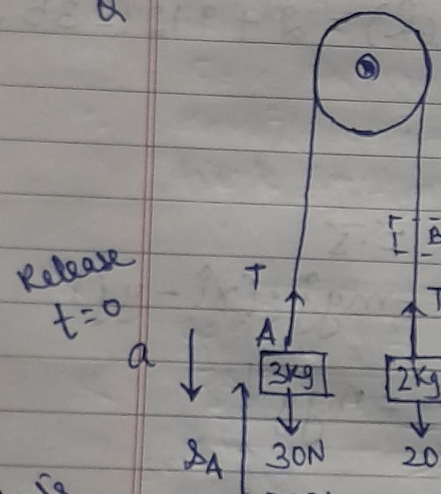


$dW_T = T\,ds\cos 90 = 0$

$\therefore \boxed{W_T = 0}$

Q



In time interval $t=0$ to $t=5s$, Find

a) $W_{gA} = mgh = 3\times10\times25 = 750$ J

b) $W_{gB} = -mgh = -2\times10\times25 = -500$ J

c) $W_{TA} = T\cdot\vec{s} = 24\times25\times\cos 180^\circ = -600$ J

d) $W_{TB} = 24\times25\times\cos 0^\circ = 600$ J

e) $W_{gnet} = W_{gA} + W_{gB} = 750 - 500 = 250$ J

f) $W_{Tnet} = W_{TA} + W_{TB} = 600 - 600 = 0$ J

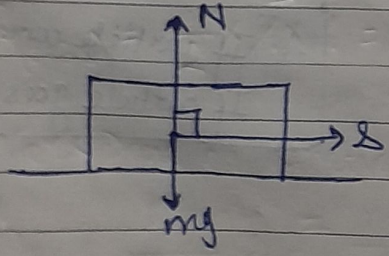
$a = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)g = \frac{3-2}{3+2}\times10 = 2\,m/s^2$ $\qquad T = \frac{2m_1m_2}{m_1+m_2}g = 24N$

$s_A = ut^0 + \frac{1}{2}at^2 = \frac{1}{2}\times2\times(5)^2 = 25m$

$s_B = 25m$

Work Done by Normal Force:

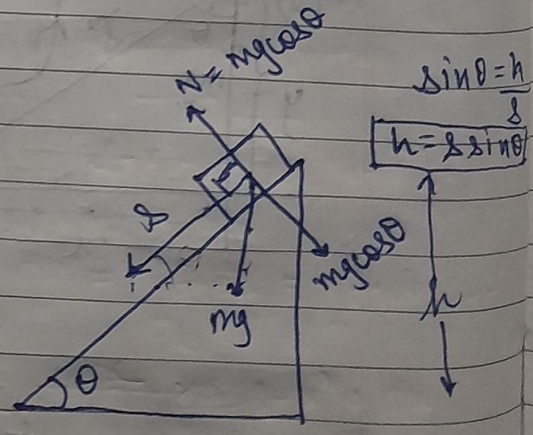
①



$W_N = N\cdot\vec{s} = N s\cos 90^\circ = 0$

$W_g = mg(s)\cos 90^\circ = 0$

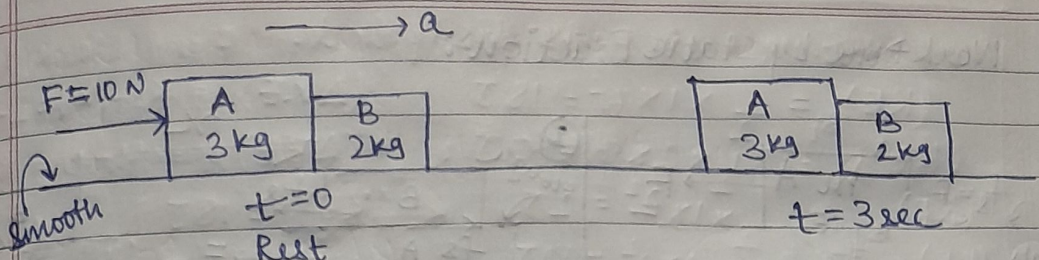
②



$W_N = N(s)\cos 90^\circ = 0$

$W_g = mg(h) = mg(s\sin\theta)$

Q



Find ① Acceleration of blocks

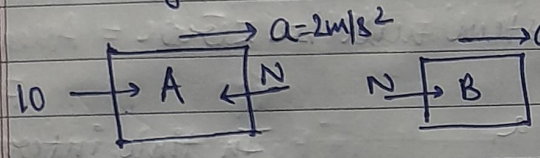
② Normal force b/w them

③ Work done by this normal on block A in $t=0$ to $t=3s$

④ Work done by this normal on block B in $t=0$ to $t=3s$

⑤ Work done by Fext on Block A and Block B

① For system: $F = ma \Rightarrow 10 = 5a \Rightarrow a = 2 m/s^2$

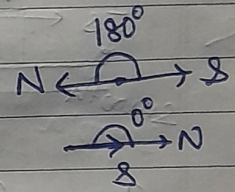
② 

$10 - N = 3 \times 2$

$N = 4$ Newton

③ $s_{t=0 \to t=3} = \frac{1}{2} \times 2 \times 3^2 = 9m$

$\therefore W_{NA} = 4 \times 9 \times \cos 180^\circ = -36J$

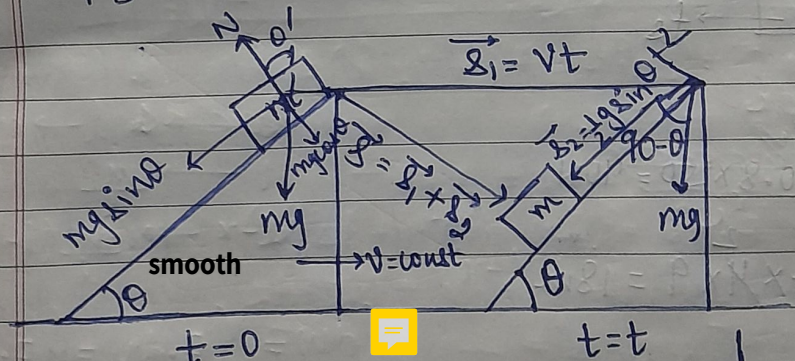


④ $W_{NB} = 4 \times 9 \times \cos 0^\circ = 36J$

⑤ $W_{FA} = 10 \times 9 \times \cos 0^\circ = 90J$

$W_{FB} = 0J$ (As F is not applied directly on block B).

Q



In this time interval ($t=0$ to $t=t$) Find W_g and W_N on block.

$W_g = m\vec{g} \cdot \vec{s} = m\vec{g} \cdot (\vec{s_1} + \vec{s_2})$

$= m\vec{g}\vec{s_1} + m\vec{g}\vec{s_2}$

$= mgs_1 \cos 90^\circ + mgs_2 \cos(90-\theta)$

$= \frac{mg^2 \sin^2\theta t^2}{2}$

$W_N = \vec{N} \cdot \vec{s} = \vec{N}(\vec{s_1} + \vec{s_2})$

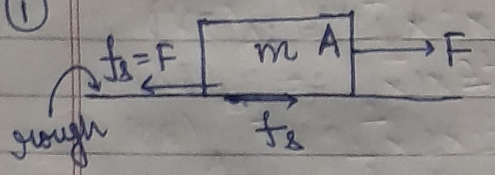
$= \vec{N}\vec{s_1} + \vec{N}\vec{s_2}$

$= Ns_1 \cos(90+\theta) + Ns_2 \cos 90^\circ$

$= -mg\cos\theta \cdot vt \sin\theta$

$= -\frac{mgut \sin 2\theta}{2}$

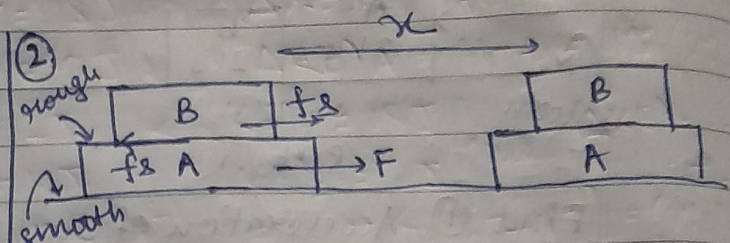
Work Done by Static Friction:

① 

$W_{f_s A} = 0$

$W_{f_s G} = 0$

$(W_{f_s})_{net} = 0$

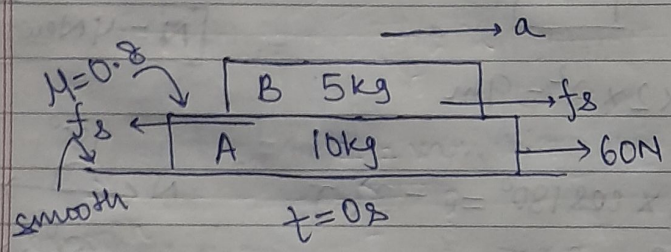
② 

$W_{f_s A} = f_s x \cos 180^\circ = -f_s x$

$W_{f_s B} = f_s x \cos 0^\circ = f_s x$

$(W_{f_s})_{net} = 0$

∴ Work done by f_s on an object can be +ve, -ve or zero.
Work done by f_s on system will be zero.

Q 

In time interval t=0 to t=3s. Find.

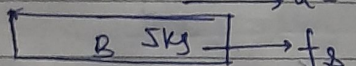
① $W_{FA} = 60 \times 18 \times \cos 0^\circ = 1080$ J

② $W_{f_A} = 20 \times 18 \times \cos 180^\circ = -360$ J

③ $W_{f_B} = 20 \times 18 \times \cos 0^\circ = 360$ J

④ $(W_f)_{net} = 0$

$a_{sys} = \frac{60}{15} = 4 m/s^2$



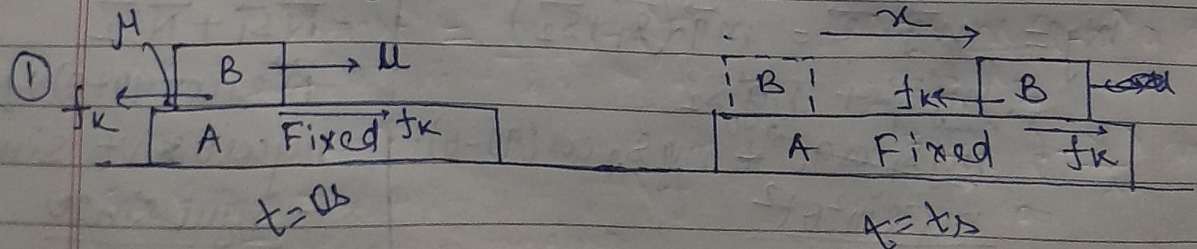
$f_s = 5 \times 4$

$= 20N$

$f_{lim} = 4N = 0.8 \times 50 = 40N$

$s_{(t=3s)} = \frac{1}{2} \times 4 \times 9 = 18m$

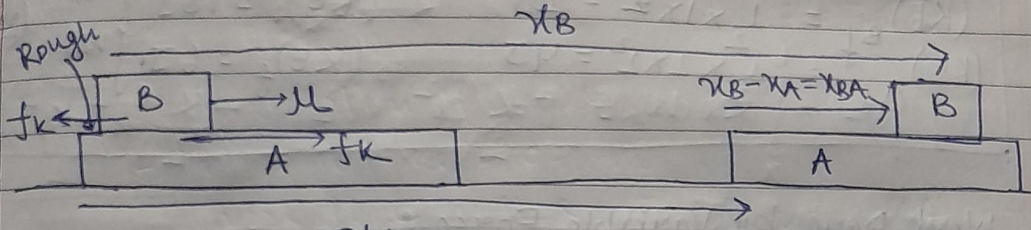
Work Done by Kinetic Friction:

① 

$W_{f_k B} = f_k x \cos 180^\circ = -f_k x$

$W_{f_k A} = f_k x 0 = 0$

$(W_{f_k})_{net} = -f_k x$

② 

$W_{f_k B} = f_k x_B \cos 180^\circ = -f_k x_B$

$W_{f_k A} = f_k x_A \cos 0^\circ = f_k x_A$

$(W_{f_k})_{net} = -f_k (x_B - x_A) = -f_k x_{BA}$ $\quad (x_B > x_A)$

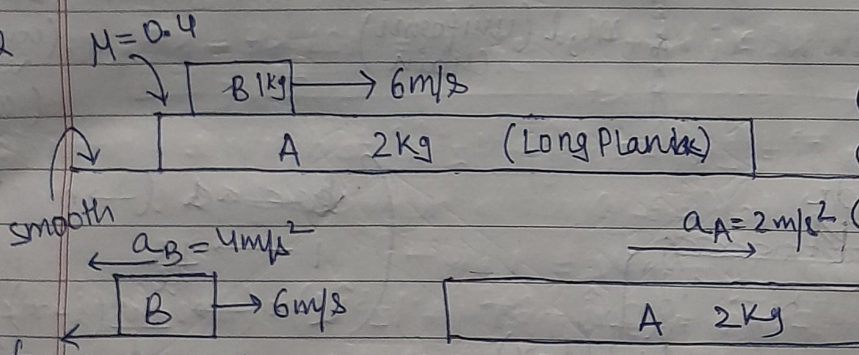
∴ Work done by f_k on an object can be +ve, −ve or zero.

→ Work done by f_k on system will be always −ve.

→ Net work done on system by f_k is $= -f_k x_{rel}$

→ This −ve W.d implies that energy of system is being dissipated in form of sound or heat or any other mean.

→ f_k is also known as dissipative frictional force.

Q 

Find W.d by f_k on

① $A = 4 \times 1 \times \cos 0^\circ = 4J$

② $B = 4 \times 2 \times \cos 180^\circ = -8J$

③ $(A+B) = -f_k x_{rel}$
$= -4 \times 3 = -12J$

$V_B = u_B - a_B t$

$V_B = 6 - 4t$

$V_A = u_A + a_A t$

$V_A = 2t$

Now, $V_A = V_B \Rightarrow 6 - 4t = 2t \Rightarrow t = 1s$

$\therefore V_B = 2 m/s \qquad V_A = 2 m/s$

$S_B \Rightarrow 2^2 - 6^2 = 2(-4)(S_B) \Rightarrow S_B = 4m$

$S_A \Rightarrow 2^2 - 0^2 = 2(2) S_A \Rightarrow S_A = 1m$

$S_{BA} = 4 - 1 = 3m$

$\therefore S_{rel} = 3m$

-ve means energy is dissipated.

$(KE)_{B_i} = \frac{1}{2} \times 1 \times 6^2 = 18\,J$ $\quad (\Delta KE)_B = K_f - K_i = 2 - 18 = -16\,J$ (W_fB)

$(KE)_{A_i} = \frac{1}{2} \times 2 \times 0 = 0\,J$ $\quad \Delta KE)_A = 4 - 0 = 4\,J$ (W_{f_A})

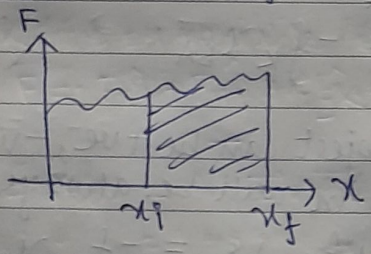
$(KE)_{B_f} = \frac{1}{2} \times 1 \times 2^2 = 2\,J$ $\quad (KE)_i = 18\,J$ $\quad (\Delta KE) = 6 - 16$

$(KE)_{A_f} = \frac{1}{2} \times 2 \times 2^2 = 4\,J$ $\quad (KE)_f = 6\,J$ net $\quad = -12\,J$

$= (W_f)_{net}$

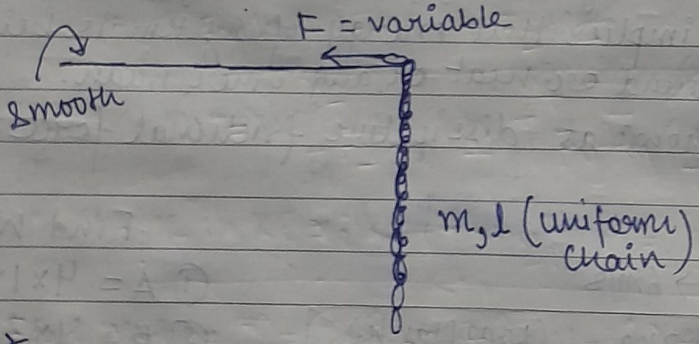
Work Done by Variable Force:

(a) F = x-axis (variable force acting along x-axis)



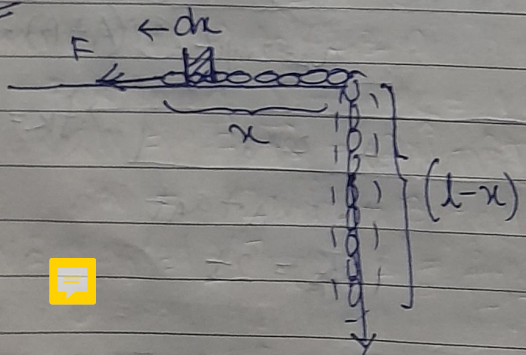
$$W = \int dW = \int_{x_i}^{x_f} F \cdot dx = \text{Area under } F\text{-}x \text{ curve}$$

Q



Find work done by external agent to pull this chain on table, slowly. (const velocity) (a=0)

After time t



$$\left(\cancel{\text{mass}}\ l \to m,\quad 1 \to \frac{m}{l},\quad l-x \to \frac{m}{l}(l-x) \right)$$

Weight of hanging part $= \frac{m}{l}(l-x)g$

$$\int dW = \int F \cdot dx = \int F\,dx\cos 0$$

$$W = \int_0^l \frac{m}{l}(l-x)g\,dx$$

$$= \frac{mg}{l}\int_0^l \left(1 - \frac{x}{l}\right) dx$$

$$W = mg\left(x - \frac{x^2}{2l}\right)_0^l = \frac{mgl}{2}$$

(For constant velocity: F = Weight of hanging part)

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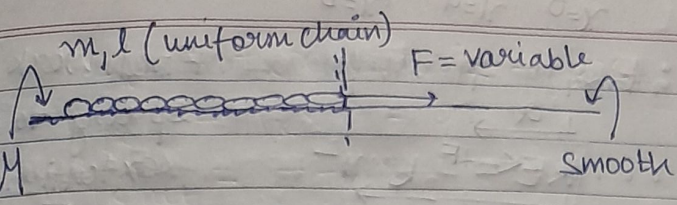
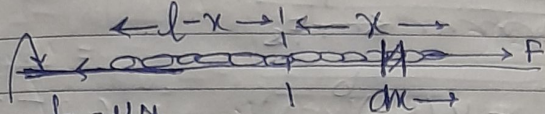
Q 

Find work done by external agent to pull this chain slowly on smooth side.



$f_k = \mu N$

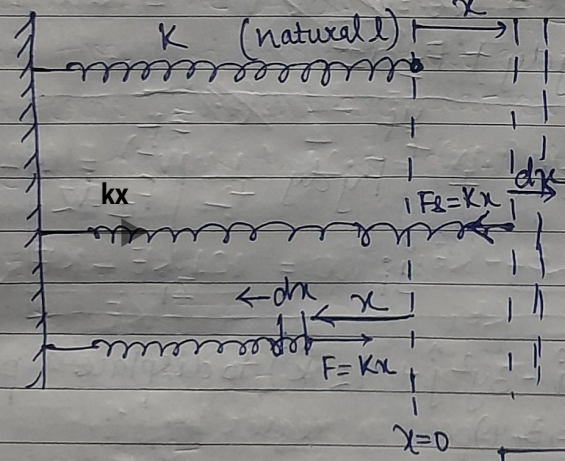
$= \mu \frac{m}{l}(l-x)g$

$F = \mu \frac{m}{l}(l-x)g$

$$\int dW = \int \vec{F}\cdot d\vec{x} = \int F\,dx\cos 0 = \int F\,dx = \int_0^l \mu \frac{m}{l}(l-x)g\,dx$$

$$W = \mu mg \int_0^l \left(1-\frac{x}{l}\right)dx = \mu mg\left(x - \frac{x^2}{2l}\right) = \frac{\mu mgl}{2}$$

Work Done by Spring Force:



$dW_s = (Kx)(dx)\cos 180°$

$$W_s = \int dW_s = -\int_{x_i}^{x_f} Kx\,dx$$

If we extend or compress from natural length to x.

$$W_s = -\int_0^x Kx\,dx = -K\left(\frac{x^2}{2}\right)_0^x$$

$$W_s = -\frac{1}{2}Kx^2\Big|_0^x$$

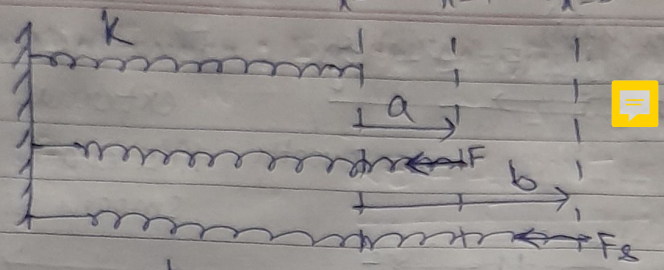
Result:

* $x=0 \to x$: extention
  $x=0 \to x$: compression
  $\Big\} \; W_s = -\frac{1}{2}Kx^2$

* x: extention $\to x=0$
  x: compression $\to x=0$
  $\Big\} \; W_s = +\frac{1}{2}Kx^2$

$$W_s = \int dW_s = -\int_{x_i}^{x_f} Kx\,dx$$

Q



Find work done by spring to increase extention from a to b

$$W = -\int_a^b kx\,dx = -\frac{k}{2}\left(x^2\right)_a^b = -\frac{k}{2}\left(b^2-a^2\right)$$

* Note: $\boxed{W_s = U_i - U_f = \frac{1}{2}kx_i^2 - \frac{1}{2}kx_f^2}$

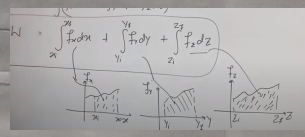
Work done by variable force in x, y, z direction:

Let $\vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$

$$W = \int dw = \int_i^f \vec{F}\cdot\vec{ds} = \int (F_x\hat{i} + F_y\hat{j} + F_z\hat{k})\cdot(dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$= \int F_x dx + F_y dy + F_z dz$$

$$\therefore \boxed{W = \int_{x_i}^{x_f} F_x dx + \int_{y_i}^{y_f} F_y dy + \int_{z_i}^{z_f} F_z dz}$$



Q Find W.d by $\vec{F} = (2x\hat{i} - 3y^2\hat{j} + 4z^3\hat{k})$ to displace a particle from (1,2,3) to (2,3,4)

$$dw = \vec{F}\cdot\vec{ds} = (2x\hat{i} - 3y^2\hat{j} + 4z^3\hat{k})\cdot(dx\hat{i} + dy\hat{j} + dz\hat{k})$$

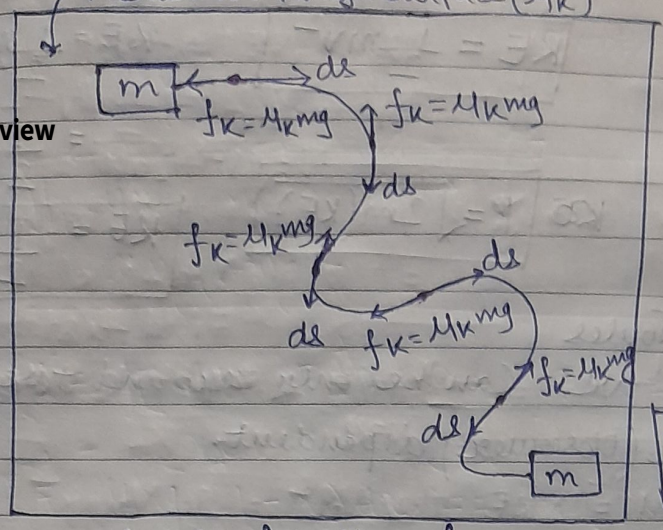
$$dw = 2x\,dx - 3y^2\,dy + 4z^3\,dz$$

$$W = \int dw = 2\int_1^2 x\,dx - 3\int_2^3 y^2\,dy + 4\int_3^4 z^3\,dz$$

$$W = 2\times\left(\frac{x^2}{2}\right)_1^2 - 3\times\left(\frac{y^3}{3}\right)_2^3 + 4\left(\frac{z^4}{4}\right)_3^4$$

$$W = 3 - 19 + 175 = 159\,J$$

Work Done by Kinetic friction if direction is changing :



$$W = \int \vec{F} \cdot \vec{ds}$$

$$= \int \vec{f_k} \cdot \vec{ds} \quad \vec{f_k} \neq \text{uniform}$$

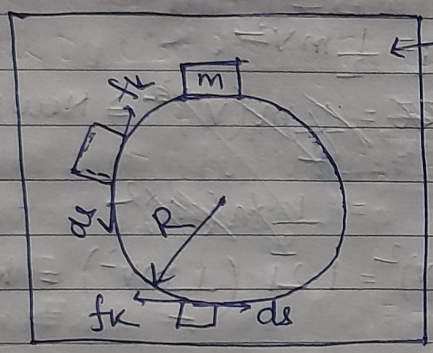
$$= \int f_k ds \cos 180^\circ$$

$$= -f_k \int ds \quad f_k = \text{uniform}$$

$$W_{f_k} = -f_k l$$

l = length of the path

Q



Find $(W_{f_k})_{Block}$

$$W_{f_k} = -f_k l$$

$$W_{f_k} = -\mu_k mg(2\pi R)$$

Energy : Capacity to do work is known as energy.

Ex Kinetic Energy, Potential Energy, Sound energy, light energy etc.
(Mechanical Energy)

→ Unit = Joules

$1 erg = 10^{-7} J$

→ Dimension = $[ML^2T^{-2}]$

→ Scalar

* Total energy of an isolated system is always conserved.
It can neither be created nor destroyed, it can only convert from one form to another.

Total Energy of an isolated system is always conserved.
Total Energy can't be created or destroyed.
Energy will convert from one form to another.

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Kinetic Energy : Energy by virtue of motion

$m \longrightarrow v$

$$KE = \frac{1}{2}mv^2$$

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}\frac{(mv)^2}{m}$$

$$P = \sqrt{2m(KE)}$$

$$KE = \frac{1}{2}\frac{P^2}{m} \qquad P = mv$$

→ Scalar, unit = Joules
→ KE cannot be −ve. Can be only zero and +ve.
→ Reference Frame (Observer) dependent.

↳ is energy by
→ Scalar
→ unit: $kg\frac{m^2}{sec^2}$ = Joule (m, $\longrightarrow v$)
→ non-negative
→ Rest ⇒ K.E. = 0
→ Ref frame dependent - Observer dependent

Q Find % change in KE if speed ↑ by 10%.

For, $v \longrightarrow KE_i = \frac{1}{2}mv^2$

New, $v' = v + \frac{10}{100}v$; ~~$KE = \frac{1}{2}m(v')^2$~~ ⇒ New speed $= v' = 1.1v$

$= \frac{1}{2}m(1.1v)^2$

For $v' \longrightarrow KE_f = \frac{1}{2}m(1.1v)^2 = 1.21\left(\frac{1}{2}mv^2\right) = 1.21(KE_i)$

$$\% \text{ change in } KE = \frac{KE_f - KE_i}{KE_i} \times 100 = \frac{1.21\,KE - KE}{KE} \times 100 = 21\%$$

Q Find % change in KE if speed increases by 1%.

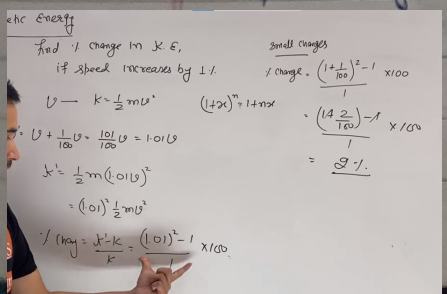
For speed $v \longrightarrow K = \frac{1}{2}mv^2$

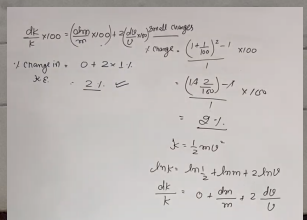
New speed $v' = v + \frac{1}{100}v = 1.01v$

For speed $v' \longrightarrow K' = \frac{1}{2}m(1.01v)^2 = 1.02\left(\frac{1}{2}mv^2\right) = 1.02K$

$$\% \text{ change in } KE = \frac{K' - K}{K} \times 100 = \frac{1.02K - K}{K} \times 100 = 2\%$$

Q





Work Energy Theorem (WET)

$\vec{F}_{net} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$

$$W = \int \vec{F}\cdot d\vec{s} = \int (F_x\hat{i} + F_y\hat{j} + F_z\hat{k})\cdot(dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$W = \int F_x dx + \int F_y dy + \int F_z dz$$

$$W = \int ma_x dx + \int ma_y dy + \int ma_z dz \qquad [F = ma]$$

$$W = \int_{v_{xi}}^{v_{xf}} m v_x \frac{dv_x}{dx}\cdot dx + \int_{v_{yi}}^{v_{yf}} m v_y \frac{dv_y}{dy} dy + \int_{v_{zi}}^{v_{zf}} m v_z \frac{dv_z}{dz} dz \qquad \left[a = v\frac{dv}{dx}\right]$$

$$W = m\int_{v_{xi}}^{v_{xf}} v_x dv_x + m\int_{v_{yi}}^{v_{yf}} v_y dv_y + m\int_{v_{zi}}^{v_{zf}} v_z dv_z$$

$$W = \frac{m}{2}\left(v_x^2\right)_{v_{xi}}^{v_{xf}} + \frac{m}{2}\left(v_y^2\right)_{v_{yi}}^{v_{yf}} + \frac{m}{2}\left(v_z\right)_{v_{zi}}^{v_{zf}}$$

$$W = \frac{m}{2}\left(v_{xf}^2 - v_{xi}^2\right) + \frac{m}{2}\left(v_{yf}^2 - v_{yi}^2\right) + \frac{m}{2}\left(v_{zf}^2 - v_{zi}^2\right)$$

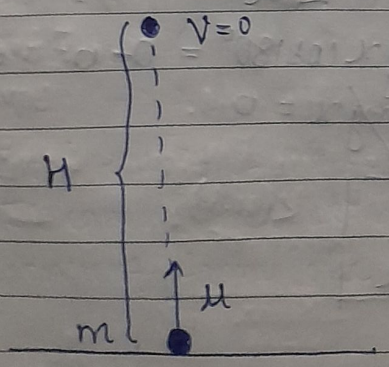
$$W = \frac{1}{2}m\left(v_{xf}^2 + v_{yf}^2 + v_{zf}^2\right) - \frac{1}{2}m\left(v_{xi}^2 + v_{yi}^2 + v_{zi}^2\right)$$

$$W = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

$$\boxed{W = (KE)_f - (KE)_i} \qquad \boxed{W_{net} = W_{All\ forces} = \Delta KE}$$

→ Work done by all the forces or work done by net force is always equal to change in K.E. ⇒ Work Energy Theorem

Q



Find Max Height achieved by the ball.

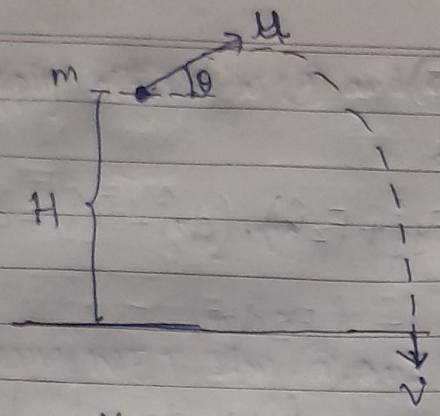
Using WET, $W_{all} = K_f - K_i$

$\Rightarrow W_g = 0 - \frac{1}{2}mu^2$

$= -mgH = -\frac{1}{2}mu^2$

$$\boxed{H = \frac{u^2}{2g}}$$

Q



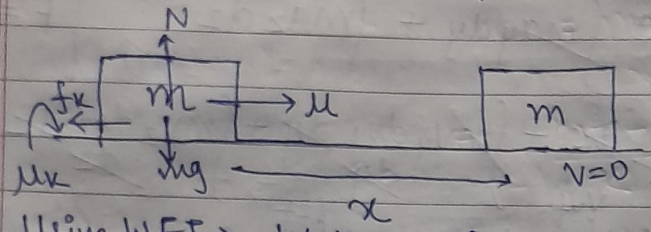
Find speed of ball when it is about to strike the ground.

Using WET ⇒ Wall = ΔKE

$$mgH = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$v = \sqrt{u^2 + 2gH}$$

Q



Find distance travelled by block before it stops

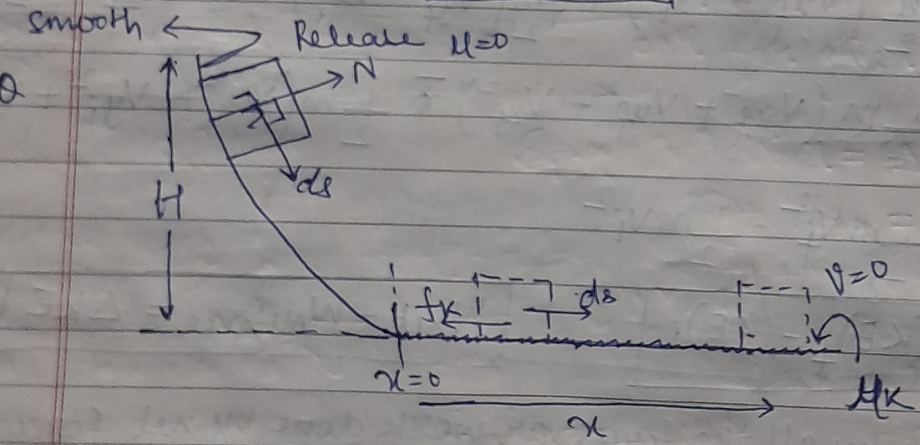
Using WET ⇒ Wall = ΔKE

$$W_g + W_N + W_{fk} = K_f - K_i$$

$$\mu_k mg \times \cos 180^\circ = -\frac{1}{2}mu^2$$

$$\boxed{x = \frac{u^2}{2\mu_k g}}$$

Q



Find distance travelled by block on horizontal surface before it stops.

Using WET ⇒ Wall = ΔKE

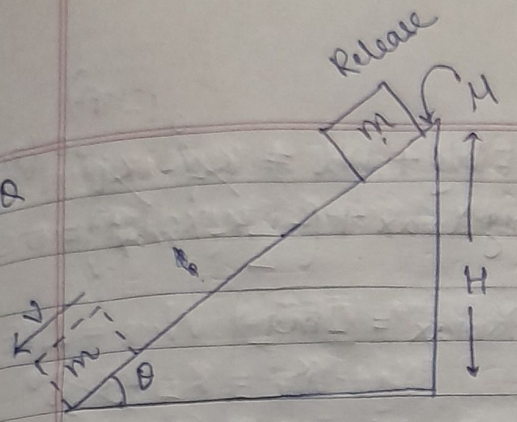
$$W_g + W_N + W_{fk} = \Delta KE$$

$$mgH + 0 + \mu_k mg \times \cos 180^\circ = 0 - 0$$

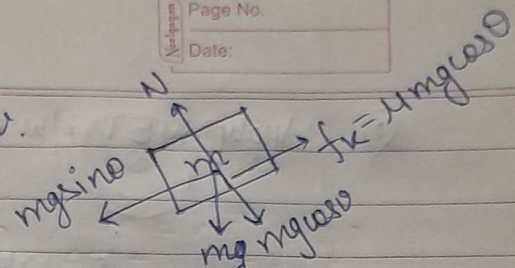
$$mgH - \mu_k mgx = 0$$

$$\boxed{x = \frac{H}{\mu_k}}$$

Q



Find v.



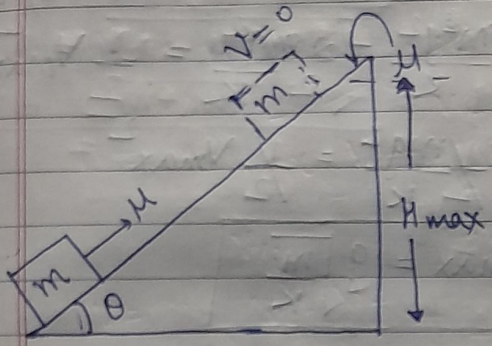
Using WET, Wall = ΔKE

$W_g + W_N + W_f = K_f - K_i$

$mgH - \mu mg\cos\theta\left(\frac{H}{\sin\theta}\right) = \frac{1}{2}mv^2$

$$v = \sqrt{2gH - 2\mu g\cot\theta H} = \sqrt{2gH(1-\mu\cot\theta)}$$

Q



Find Hmax

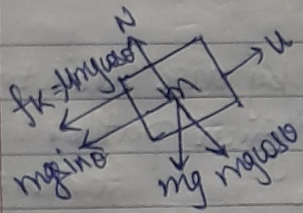
Using WET ⇒ Wall = ΔKE

$W_g + W_N + W_f = K_f - K_i$

$-mgH_{max} - \mu mg\cos\theta\left(\frac{H_{max}}{\sin\theta}\right) = -\frac{1}{2}mu^2$

$$H_{max}\left(g + \mu g\cot\theta\right) = \frac{1}{2}u^2$$

$$\boxed{H_{max} = \frac{u^2}{2(g+\mu g\cot\theta)}}$$



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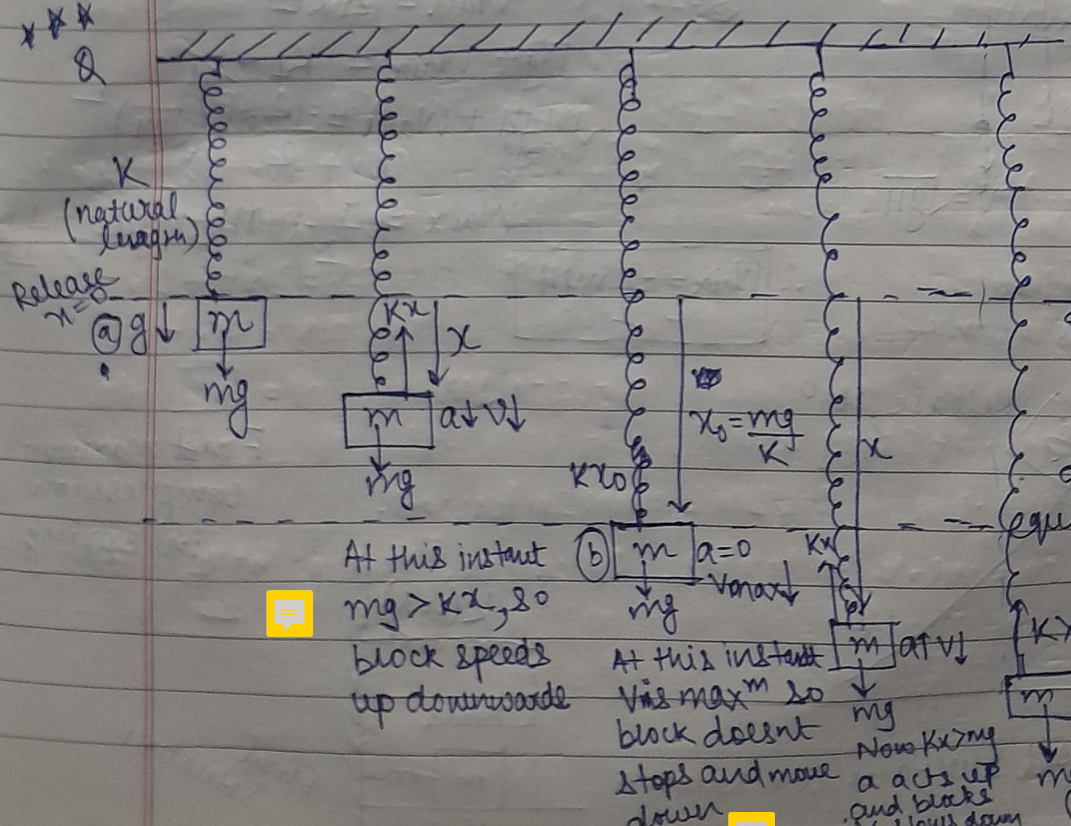
Q



① Find maximum extension in spring when block is released

② Also find Vmax

① Apply WET b/w (a) and (c) ⇒ $W_g + W_s = K_f - K_i$ (Stops, $K_f = 0$; Release, $K_i = 0$)

$= mg X_{max} + \left(-\frac{1}{2} K X_{max}^2\right) = 0$

$$\boxed{X_{max} = \frac{2mg}{K}}$$

② Using WET b/w (a) and (b) ⇒ $W_g + W_s = K_f - K_i$ ($K_i = 0$)

$= mg\left(\frac{mg}{K}\right) + -\frac{1}{2} K \left(\frac{mg}{K}\right)^2 = \frac{1}{2} m V_{max}^2$

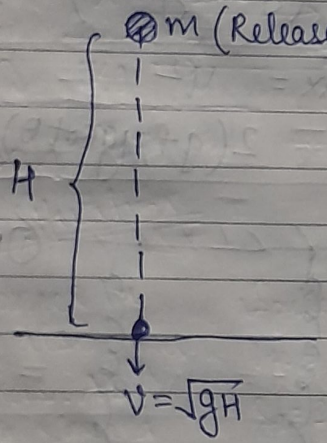
$= \frac{m^2 g^2}{K} - \frac{1}{2}\frac{mg^2}{K} = \frac{1}{2} V_{max}^2$

$= \frac{1}{2}\frac{mg^2}{K} = \frac{1}{2} V_{max}^2$

$$\boxed{V_{max} = g\sqrt{\frac{m}{K}}}$$

R-R

Q



Find W.d by air resistance on ball for given path.

$W_{all} = \Delta KE$

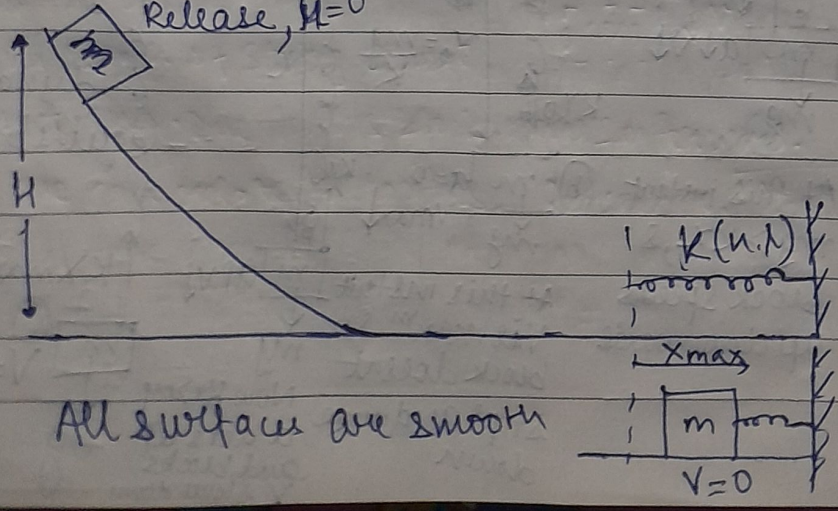
$W_g + W_{air} = K_f - K_i$ ($K_i = 0$)

$mgH + W_{air} = \frac{1}{2} m \left(\sqrt{gH}\right)^2$

$W_{air} = \frac{1}{2} mgH - mgH = -\frac{1}{2} mgH$

$$\therefore \boxed{W_{air} = -\frac{1}{2} mgH}$$

Q



Find maximum compression in spring

All surfaces are smooth

Final situation

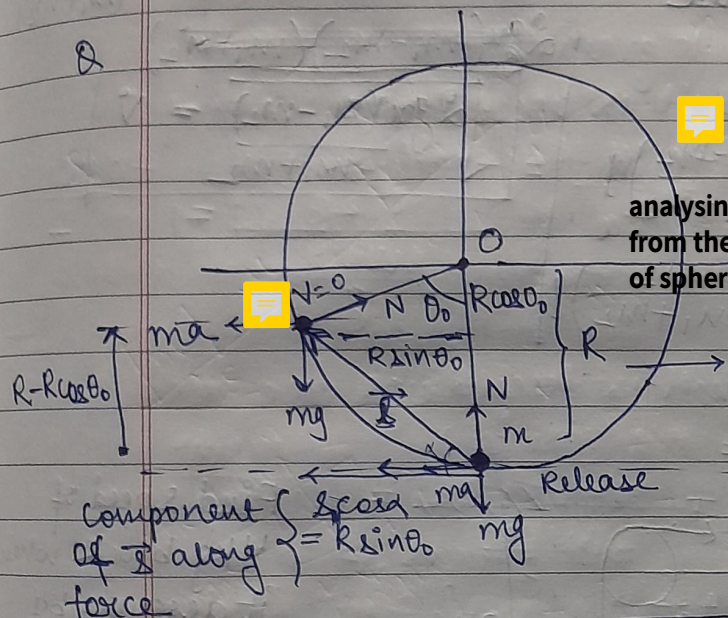
Using WET, Wall = ΔKE

$W_g + W_N + W_s = K_f - K_i$

$mgH - \frac{1}{2}kx_{max}^2 = 0$

$$\boxed{x_{max} = \sqrt{\frac{2mgH}{k}}}$$

Find maximum angle b/w line joining O and mass 'm' with vertical.



analysing from the frame of sphere

Using WET, Wall = ΔKE

$W_g + W_N + W_{Pseudo} = K_f - K_i$

$-mgR(1-\cos\theta_0) + ma(R\sin\theta_0) = 0$

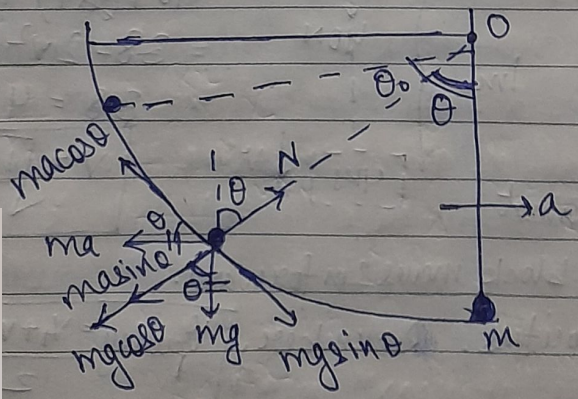
$g(1-\cos\theta_0) = a\sin\theta_0$

$g\left(2\sin^2\frac{\theta_0}{2}\right) = a\left(2\sin\frac{\theta_0}{2}\cos\frac{\theta_0}{2}\right)$

$\tan\frac{\theta_0}{2} = \frac{a}{g}$

$$\therefore \boxed{\theta_0 = 2\tan^{-1}\left(\frac{a}{g}\right)}$$

Analysis Why does the particle goes up?

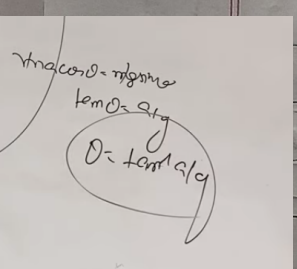


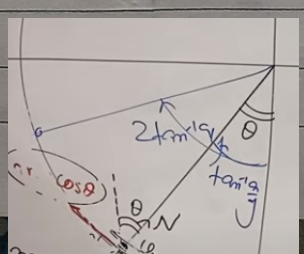
Now, as θ↑, cosθ↓ sinθ↑

initially cosθ > sinθ

$ma\cos\theta > mg\sin\theta$

So particle was moving up but slowly its acc^n up was decreasing. When macosθ becomes equal to mgsinθ it velocity becomes maximum and hence it travels further distance than θ angle and reaches max θ_0 where $mg\sin\theta > ma\cos\theta$ but V=0 (moment





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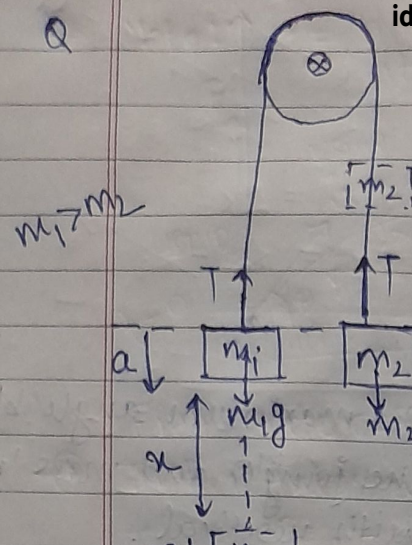


Find speed of blocks when m_1 descends by distance x.

on m1+m2 system

Using WET $\Rightarrow W_{all} = \Delta KE$

$W_g + W_T = K_f - K_i$ (0)

$m_1 gx - m_2 gx + Tx - Tx = \frac{1}{2} m v^2$ ($m_1 + m_2$)

$$v = \sqrt{\frac{2gx(m_1 - m_2)}{m_1 + m_2}}$$

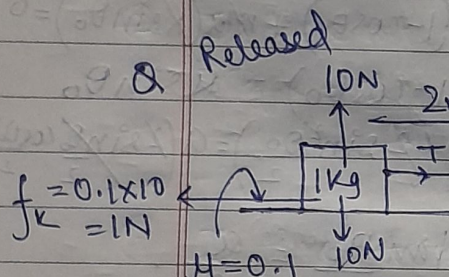
$$a = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) g$$

$$T = \left(\frac{2m_1 m_2}{m_1 + m_2}\right) g$$

$v^2 = 2\frac{(m_1 - m_2)}{m_1 + m_2} g x$ (release)

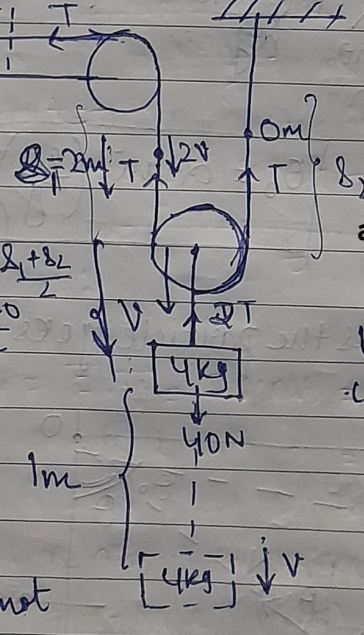
$a = \frac{dv}{dx} v = \frac{2(m_1 - m_2)g}{(m_1 + m_2)}$ (1)

Q



ideal conditions

Find speed of 4kg block when it descends by distance 1m



Explanation of situation

When block comes 1m down, pulley also comes 1m downwards. As String's length should remain same, left side string has to come 2m down as the other end is fixed and length of string cannot change on that end. Due to this block moves 2m forward.

$v_{pulley} = \frac{s_1 + s_2}{2}$

$1m = \frac{s_1 + 0}{2}$

$s_1 = 2m$

applying WET on 4kg + 1kg

$W_{all} = \Delta KE$

$W_g + W_T + W_f + W_N = K_f - K_i$ (0)

$-40 + (-2T) + 2T + (-2) = \frac{1}{2} \times v^2 +$

$38 = 2v^2 + 2v^2$

$38 = 4v^2$

$$v = \sqrt{\frac{19}{2}} \text{ m/s}$$

Apply WET on (4kg + 1kg)

$-4 \times 10 \times 1 - (2T)(1) + T(2)$

$= 0(2)$

$= \frac{1}{2} \times 4 \times v^2 + \frac{1}{2} \times 1 (2v)^2$

$38 = 2v^2 + 2v^2$

$38 = 4v^2$

$v = \sqrt{\frac{19}{2}}$ m/sec

Similarly we can relate velocities. Remember, $v_{pulley} = \frac{v_1 + v_2}{2}$ $\quad s_{pulley} = \frac{s_1 + s_2}{2}$

m_2 will move when $F > (f_{lim})_{m_2}$
m_1 will move when $Kx > (f_{lim})_{m_1}$

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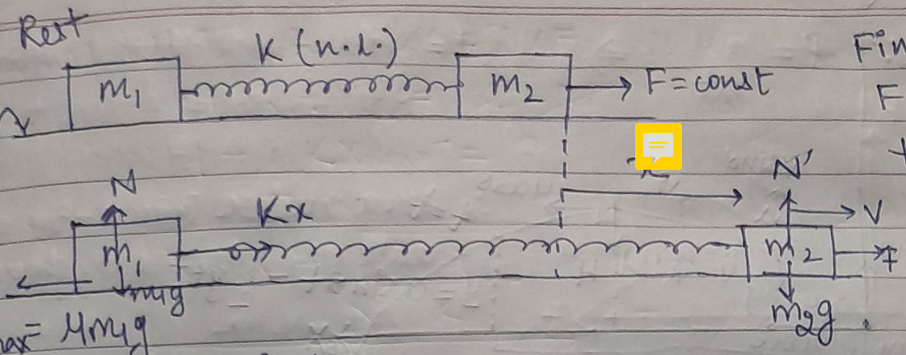
Q



Find min value of F to just move the block m_1

$f_{s\max} = \mu m_1 g$

For block m_1 to just move

$Kx = \mu m_1 g$

$$\boxed{x = \frac{\mu m_1 g}{K}}$$

Now, Apply WET on m_2

$$W_g + W_N + W_F + W_s + W_f = K_f - K_i$$

$$Fx - \frac{1}{2}kx^2 - \mu m_2 g x = \frac{1}{2}mv^2$$

Now for $\min^m F$, v should be $\min^m \Rightarrow v = 0^+$

$$Fx - \frac{1}{2}kx^2 - \mu m_2 g x = 0$$

$$F = \frac{Kx}{2} + \mu m_2 g \Rightarrow \boxed{F = \frac{\mu m_1 g}{2} + \mu m_2 g}$$

Power: Rate of work done is defined as Power.

Average Power (for time interval): $P_{avg} = \frac{\Delta W}{\Delta t} = \frac{\text{Work Done}}{\text{Time taken}}$

Instantaneous Power (for any instant): $P_{inst} = \frac{dW}{dt} = \frac{\vec{F} \cdot d\vec{s}}{dt}$

$$\therefore \boxed{P_{inst} = \frac{dW}{dt} = \vec{F} \cdot \vec{v}}$$

$= \vec{F} \cdot \vec{v}$

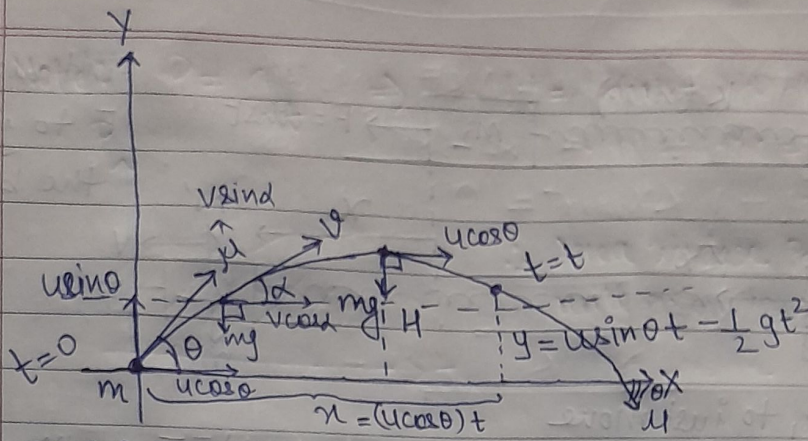
$= Fv\cos\theta$

$= Fv\cos\theta$ ↳ angle b/w $\vec{F}$ and $\vec{v}$

$= F v_{\parallel} = v F_{\parallel}$

→ Power is scalar quantity
→ Can be +ve, −ve, zero
→ Unit = Joule/sec = watt (W)
1 HorsePower = 746 W
→ Dimension = $[ML^2T^{-3}]$

Q



Find Power of force due to gravity:-

① In time interval when ball is ascending:

$$P_{avg} = \frac{\Delta W}{\Delta T} = \frac{-mgH}{t_a} = \frac{-mg\left(\frac{u^2\sin^2\theta}{2g}\right)}{\frac{u\sin\theta}{g}} = \frac{-mgu\sin\theta}{2}$$

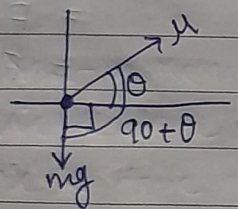
② In time interval when ball is descending:

$$P_{avg} = \frac{\Delta W}{\Delta T} = \frac{mgH}{t_d} = \frac{mg\left(\frac{u^2\sin^2\theta}{2g}\right)}{\frac{u\sin\theta}{g}} = \frac{mgu\sin\theta}{2}$$

③ In time interval t=0 to t=t:

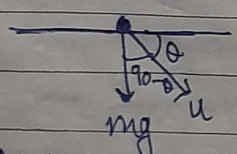
$$P_{avg} = \frac{\Delta W}{\Delta T} = \frac{-mgy}{t} = \frac{-mg\left(u\sin\theta t - \frac{1}{2}gt^2\right)}{t} = -mg\left(u\sin\theta - \frac{1}{2}gt\right)$$

④ At the time of projection:



$$P_{inst} = \vec{F}\cdot\vec{v} = Fv\cos\theta = mgu\cos(90+\theta)$$
$$= -mgu\sin\theta$$

⑤ At the time when ball is about to strike the ground:



$$P_{inst} = \vec{F}\cdot\vec{v} = mgu\cos(90-\theta)$$
$$= mgu\sin\theta$$

⑥ At the time when ball is at max^m height.

$$P_{inst} = \vec{F}\cdot\vec{v} = mg(u\cos\theta)\cos 90^\circ = 0$$

⑦ When ball is moving at an angle α with the horizontal:

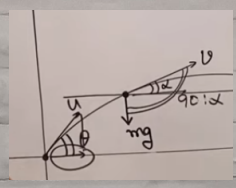
$P_{inst} = \vec{F} \cdot \vec{v} = mg(v)\cos(90+\alpha) = -mgv\sin\alpha$

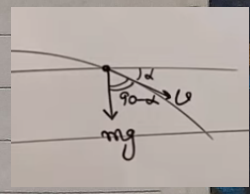
Also $v\cos\alpha = u\cos\theta$

$v = \frac{u\cos\theta}{\cos\alpha}$

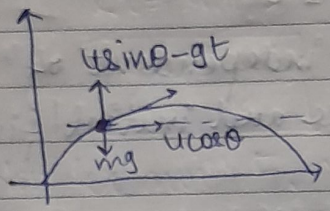
$= -mg\frac{u\cos\theta}{\cos\alpha}\sin\alpha$

$= -mgu\cos\theta\tan\alpha$





⑧ At any time 't':

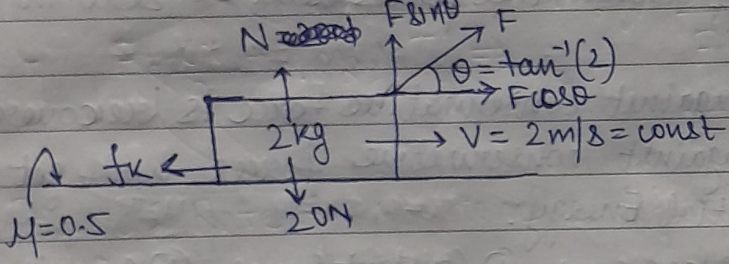


$P_{inst} = \vec{F} \cdot \vec{v}$

$= (-mg\hat{j}) \cdot (v_x\hat{i} + v_y\hat{j})$

$= -mgv_y = -mg(u\sin\theta - gt)$

Q



① Find value of F

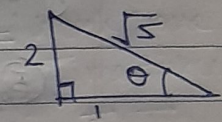
② Find Power of
a) mg b) N
c) f_k d) F

① $N + F\sin\theta = 20 \Rightarrow N = 20 - F\sin\theta$

$F\cos\theta = f_k = \mu N = 0.5(20 - F\sin\theta)$

$F\cos\theta = 10 - \frac{F\sin\theta}{2}$

$F\left(\cos\theta + \frac{\sin\theta}{2}\right) = 10 \Rightarrow F = \frac{20}{2\cos\theta + \sin\theta} = \frac{20}{2\left(\frac{1}{\sqrt{5}}\right) + \frac{2}{\sqrt{5}}}$



$F = \frac{20}{4/\sqrt{5}} = 5\sqrt{5}\ N$

② a) $mgv\cos 90^\circ = 0 \Rightarrow P_{mg} = 0$

b) $Nv\cos 90^\circ = 0 \Rightarrow P_N = 0$

c) $f_k v\cos 180^\circ = -f_k v = -5 \times 2 = -10W$

d) $Fv\cos(\theta) = +Fv\cos\theta = +5\sqrt{5} \times 2 \times \frac{2}{\sqrt{5}} = +20N$

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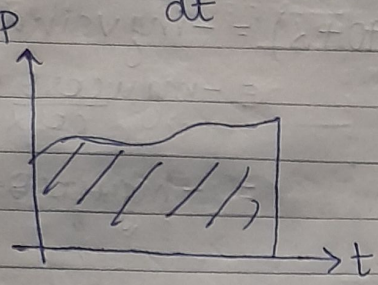
* Note: $P = \frac{dW}{dt} \Rightarrow W = \int dW = \int P\,dt$



∴ Area under Power–time curve will provide work done.

Conservative Forces:

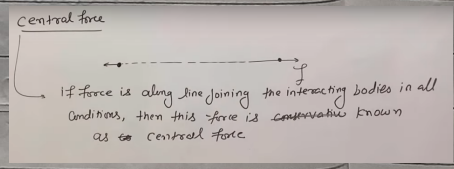
→ Work Done by conservative forces is independent of path followed and depends only on force and initial/final position.

→ Work Done by conservative forces will be zero in a closed path.

→ Work done against conservative force is recoverable.

→ Work done against conservative forces will get stored in form of Potential Energy.

→ Eg of conservative forces → ① Constant forces, ② Gravitational forces b/w charges ③ Electrostatic force, ④ magnetic force b/w magnets, ⑤ spring force / elastic force, ⑥ Central force → If force is along the line joining the interacting bodies in all conditions, then this force is known as central force.



Non-conservative forces:

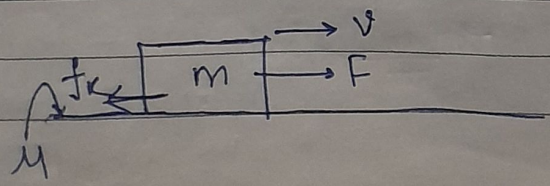
→ Work done is path dependent.

→ Work done along closed path may not be zero

→ Eg- ① Kinetic friction, ② Air resistance, ③ viscous force etc

→ Any work done against non-conservative forces cannot be stored, rather it dissipates in form of sound or heat or any other form of energy. Thus, PE is not defined for non-conservative forces

Ex



Energy here dissipates in form of sound and heat.

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Q Find W.d. by $\vec{F} = 2x\hat{i} + 3y^2\hat{j}$ to displace a particle from $(0,0)$ to $(2,3)$

$$W = \int dW = \int \vec{F}\cdot d\vec{s} = \int (2x\hat{i} + 3y^2\hat{j})\cdot(dx\hat{i} + dy\hat{j}) = \int_0^2 2x\,dx + \int_0^3 3y^2\,dy$$

$$= \frac{2}{2}(x^2)_0^2 + \frac{3}{3}(y^3)_0^3 = 4 + 27 = 31\,J$$

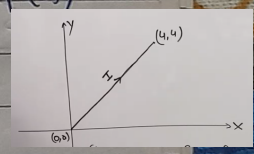
Is this force conservative or not?
Yes, it is conservative as we dont need the idea of path followed

Q Find W.d. by $\vec{F} = x\hat{i} + xy\hat{j}$ to displace a particle from $(0,0)$ to $(4,4)$. Is this $\vec{F}$ conservative or non-conservative

$$W = \int dW = \int \vec{F}\cdot d\vec{s} = \int (x\hat{i} + xy\hat{j})\cdot(dx\hat{i} + dy\hat{j}) = \int_0^4 x\,dx + \int_{(0,0)}^{(4,4)} xy\,dy$$

$$W = \left(\frac{x^2}{2}\right)_0^4 + \int_{(0,0)}^{(4,4)} xy\,dy = 8 + \int_{(0,0)}^{(4,4)} xy\,dy$$

Here, we cannot find w.d until we have $y = f(x)$ or $x = f(y)$ ie eqn of path. So this force is non-conservative.



(I) Now, if it is given path followed is a st. line $y = x$

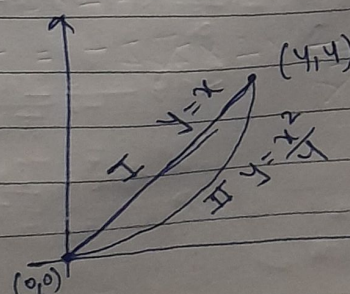
$$W = 8 + \int xy\,dy = 8 + \int_0^4 y^2\,dy = 8 + \left(\frac{y^3}{3}\right)_0^4 = 8 + \frac{64}{3} = \frac{88}{3}\,J$$

(II) If given path is parabola $y = \frac{x^2}{4}$

$$W = 8 + \int xy\,dy = 8 + \frac{1}{4}\int_0^4 x^3\left(\frac{x}{2}\right)dx = 8 + \frac{1}{8}\int_0^4 x^4\,dx = 8 + \frac{1024}{40} = \frac{168}{5}\,J$$

$y = \frac{x^2}{4}$

$dy = \frac{x}{2}dx$



W.d. along path II

$$W = \int dw = \int_0^4 x\,dx + \int xy\,dy$$

$$= 8 + 2\int_0^4 y^{3/2}\,dy$$

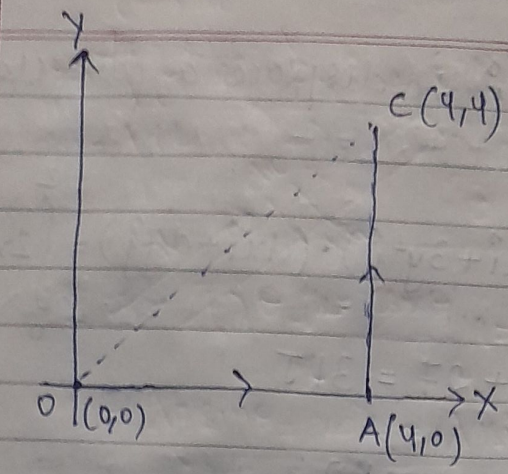
$$= 8 + 2\cdot\frac{(y^{5/2})_0^4}{5/2}$$

$$= 8 + \frac{4}{5}\cdot(4^{5/2} - 0^{5/2})$$

$$= 8 + \frac{4}{5}\cdot(2^5 - 0)$$

$$= 8 + \frac{4}{5}\times 32$$

(III)



Find W.d along path OAC

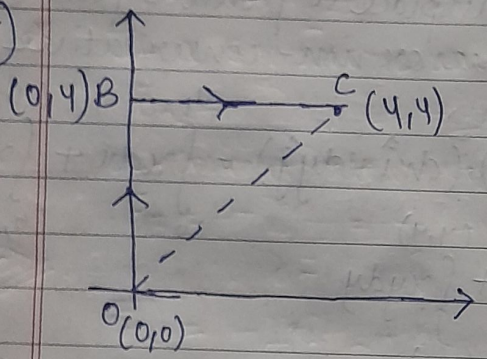
$W = \int dw = \int x\,dx + \int xy\,dy$

W_{OA}	W_{AC}
$y=0,\ dy=0$	$x=4,\ dx=0$
$W = \int_0^4 x\,dx$	$W = \int_0^4 4y\,dy$
$W = 8\,J$	$W = 4\int_0^4 y\,dy$
	$= 4\times 8 = 32\,J$

$W_{OAC} = W_{OA} + W_{AC}$

$= 8 + 32 = 40\,J$

(IV)



Find W.d along path OBC

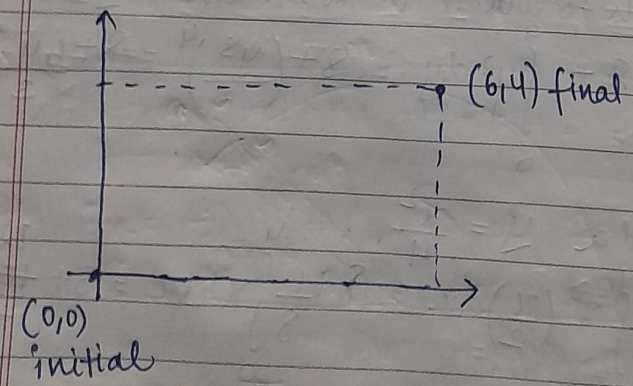
$W = \int dw = \int x\,dx + \int xy\,dy$

W_{OB}	W_{BC}
$x=0, dx=0$	$y=4,\ dy=0$
$W=0$	$W = \int_0^4 x\,dx$
	$W = 8\,J$

$W_{OBC} = W_{OB} + W_{BC}$

$= 0 + 8 = 8\,J$

Q



Find W.d. by $\vec{F} = y\hat{i} + x\hat{j}$ to displace a particle from (0,0) to (6,4)

$\int dw = \int \vec{F}\cdot d\vec{s}$

$W = \int (y\hat{i} + x\hat{j})\cdot(dx\hat{i} + dy\hat{j})$

$W = \int y\,dx + x\,dy = \int d(xy)$

$W = (xy)_{(0,0)}^{(6,4)} = 24\,J$

tihs force is conservative in nature, since we did not need path to calculate work done

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Potential Energy:

→ It is energy by virtue of relative position, orientation or configuration.

→ It is defined for conservative forces only.

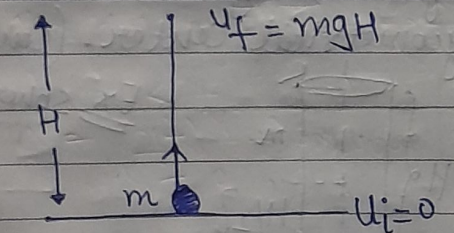
→ Any work done against conservative forces will get stored in form of potential energy.

★ **Change in Potential Energy = −W.d by Conservative forces**

$$\boxed{dU = -dW_c}$$

$$\boxed{\Delta U = -W_c}$$

Gravitational Potential Energy:



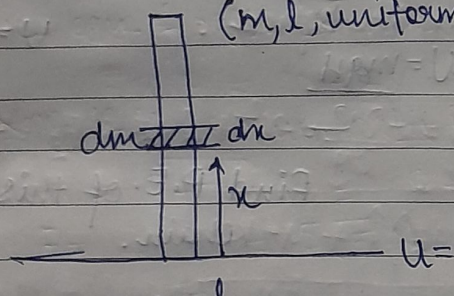
$\Delta U = -W_g$

$U_f - U_i = -(-mgH)$

$U_f - U_i = mgH$

$$\boxed{U_f = U_i + mgH}$$

Q (m, l, uniform rod) Find G.P.E. of rod.



$\int dU = \int dm\,g\,x$

$= \int dU = \int \lambda g x\,dx$

$\Rightarrow U = \lambda g \int_0^l x\,dx = \frac{\lambda g}{2}(x^2)_0^l = \frac{\lambda g}{2} \times l^2$

$= \frac{m}{l} \times \frac{1}{2} \times g l^2 = \frac{mgl}{2}$

$\therefore \boxed{U = \frac{mgl}{2}}$

mass per unit length

$l \longrightarrow m$

$1 \longrightarrow \frac{m}{l} = \lambda$

~~dx = dx~~

$m = \lambda l$

for dx length mass = dm

$dm = \lambda\,dx$

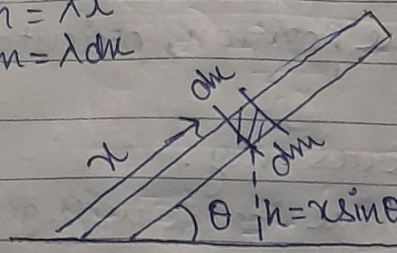
$l \rightarrow m$

$dx \rightarrow \frac{m}{l} = \lambda$

Q (m, l, uniform rod) Find G.P.E of Rod.

$m = \lambda l$

$dm = \lambda dx$



$\int dW = \int dm\, g\, x \sin\theta$

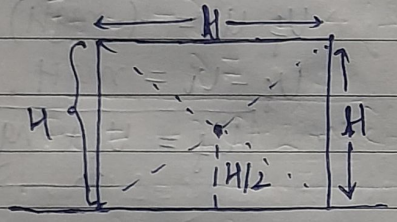
$U = \int_0^l \lambda\, dx\, g\, x \sin\theta$

$U = \lambda g \sin\theta \int_0^l x\, dx = \lambda g \sin\theta \frac{l^2}{2} = \frac{m}{l} \times g \sin\theta \frac{l^2}{2}$

$$\boxed{U = \frac{mgl\sin\theta}{2}}$$

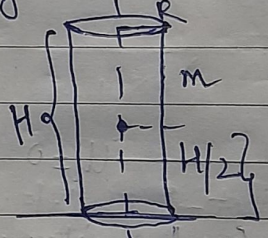
* If a symmetric uniform object is given then its potential energy can be written by assuming its entire mass to be concentrated at its geometrical center. (Center of mass)

Ex



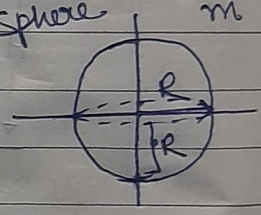
$U = \frac{mgH}{2}$

Ex



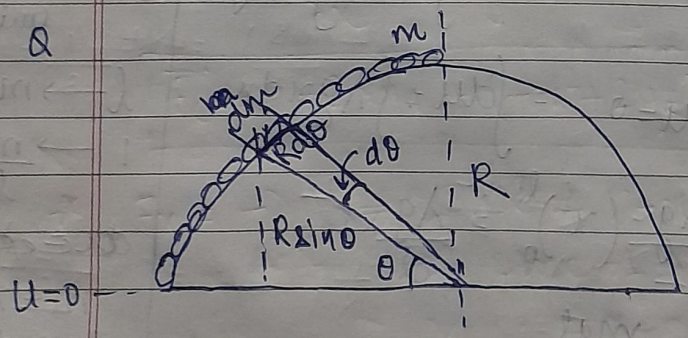
$U = \frac{mgH}{2}$

Ex sphere



$U = mgR$

Q



Find P.E. of this uniform chain.

$\frac{\pi}{2} R \rightarrow m$

$\lambda = \frac{m}{\frac{\pi}{2} R}$

$R d\theta \rightarrow \frac{2m}{\pi R} \times R d\theta$

$\lambda = \frac{dm}{Rd\theta}$

$dm = \frac{2m}{\pi} d\theta$

$Rd\theta\lambda = dm$

$Rd\theta \frac{m}{\frac{\pi}{2}R} = dm$

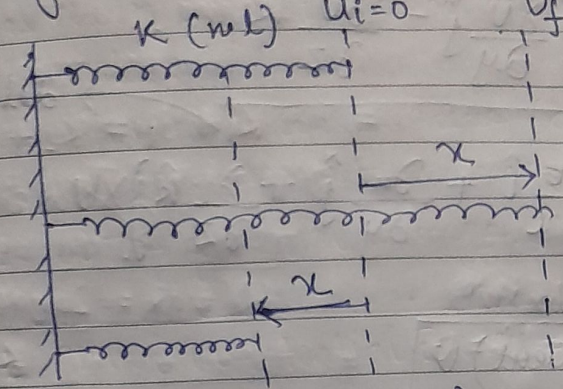
$\int dU = \int dm\, g (R\sin\theta)$

$U = \int \frac{2m}{\pi} d\theta\, g\, R\sin\theta$

$U = \frac{2mgR}{\pi} \int_0^{\pi/2} \sin\theta\, d\theta$

$$\boxed{U = \frac{2mgR}{\pi}}$$

Spring Potential Energy :



$U_f = U_i + \frac{1}{2}kx^2$

$= \frac{1}{2}kx^2$

$\Delta U = -W_s$

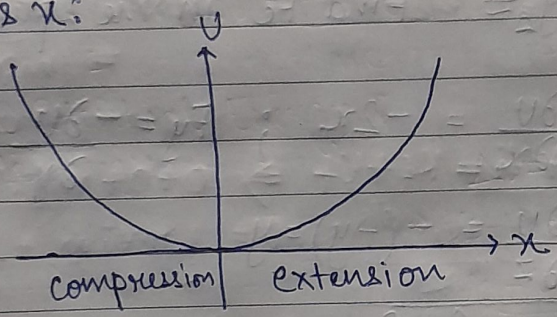
$U_f - U_i = -\left(-\frac{1}{2}kx^2\right)$

$$\boxed{U_f = U_i + \frac{1}{2}kx^2}$$

$U_f = U_i + \frac{1}{2}kx^2$

$= \frac{1}{2}kx^2$

Graph of U v/s x :



General Relation b/w Potential energy and conservative force :

$$dU = -dW_c$$

$$= -\vec{F_c} \cdot \vec{ds}$$

$$= -(F_x\hat{i} + F_y\hat{j} + F_z\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$dU = -F_x dx - F_y dy - F_z dz$$

If y, z are constant $\Rightarrow dy = dz = 0$

$$dU = -F_x dx$$

$$\boxed{F_x = \left(\frac{dU}{dx}\right)_{y,z=\text{const}}}$$

partial differentiation

$$\therefore \boxed{F_x = -\frac{\partial U}{\partial x}}$$

$$\boxed{F_y = -\frac{\partial U}{\partial y}} \quad (x, z \text{ are const})$$

$$\boxed{F_z = -\frac{\partial U}{\partial z}} \quad (x, y \text{ are const})$$

$\Rightarrow \vec{F_c} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$

$$\boxed{\vec{F_c} = -\frac{\partial U}{\partial x}\hat{i} - \frac{\partial U}{\partial y}\hat{j} - \frac{\partial U}{\partial z}\hat{k}}$$

→ del operator

$$\vec{F_c} = -\left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right) U$$

nabla symbol

$$= -\nabla U$$

$$\boxed{\vec{F_c} = -(\text{gradient})\, U}$$

Q $U = x^2 - yz$. Find $\vec{F_c}$ in the space.

$$F_x = -\frac{\partial U}{\partial x} = -2x \; ; \; F_y = -\frac{\partial U}{\partial y} = -(-z) = z$$

$$F_z = -\frac{\partial U}{\partial z} = -(-y) = y$$

$$\boxed{\vec{F_c} = -2x\hat{i} + z\hat{j} + y\hat{k}}$$

Q $U = x^2y - y^2z$. Find $\vec{F_c}$ in the space.

$$F_x = -\frac{\partial U}{\partial x} = -2xy \; ; \; F_y = -\frac{\partial U}{\partial y} = -x^2 + 2yz \; ; \; F_z = -\frac{\partial U}{\partial z} = y^2$$

$$\boxed{\vec{F_c} = -2xy\hat{i} + (-x^2 + 2yz)\hat{j} + y^2\hat{k}}$$

* $F_x = -\frac{\partial U}{\partial x} \Rightarrow$ Slope of U-x curve $= \frac{\partial U}{\partial x}$, and negative of slope of U-x curve will give F_x.

Q $\vec{F_c} = 2x\hat{i} - 3y^2\hat{j} - 4z^3\hat{k}$. If U at $(0,0,0)$ is 5J. Find U at (x,y,z)

$$dU = -dW_c$$

$$dU = -(2x\hat{i} - 3y^2\hat{j} - 4z^3\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$\int_5^U dU = -\int_0^x 2x\,dx + \int_0^y 3y^2\,dy + \int_0^z 4z^3\,dz$$

$= U-5 = -(x^2)_0^x + (y^3)_0^y + (z^4)_0^z$

$= U = -x^2+y^3+z^4+5$

Total Mechanical Energy Conservation (T.M.E.C):

Total Mechanical Energy = K.E + P.E.

According to W.E.T $\Rightarrow$ Wall = ΔKE

$= W_{int} + W_{ext} = \Delta KE$

$= W_C + W_{NC} + W_{ext} = \Delta KE$

$= -\Delta U + W_{NC} + W_{ext} = \Delta KE$

$= W_{NC} + W_{ext} = \Delta KE + \Delta U$

$= W_{NC} + W_{ext} = \Delta(TME)$

If $W_{NC} = W_{ext} = 0 \Rightarrow \Delta(TME) = 0$

$(TME)_i = (TME)_f$

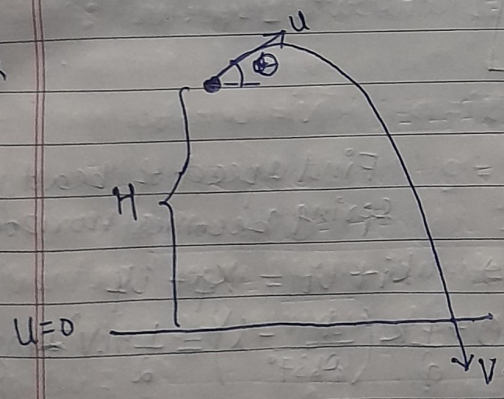
$\boxed{KE_i + U_i = KE_f + U_f}$

$= U_i - U_f = KE_f - KE_i$

$= \boxed{\text{Loss in PE} = \text{Gain in KE}}$

$\therefore$ If W_{NC} and W_{ext} is zero on an system then total mechanical energy of system is conserved or constant.

Q



Find V.

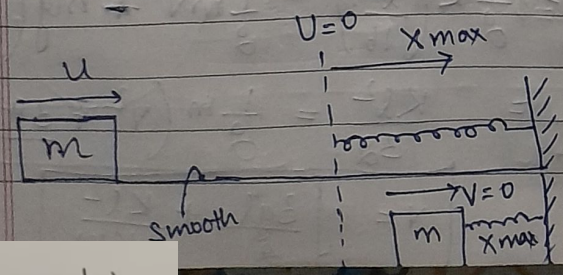
$\Rightarrow$ Here, $W_{ext} = W_{NC} = 0$

$= KE_i + U_i = KE_f + U_f$

$= \frac{1}{2}mu^2 + mgH = \frac{1}{2}mv^2 + 0$

$\boxed{v = \sqrt{u^2 + 2gH}}$

Q



Find max compression in spring

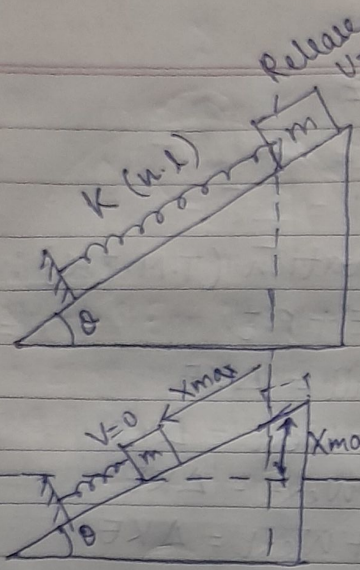
$\Rightarrow K_i + U_i = K_f + U_f$

$= \frac{1}{2}mu^2 + 0 = 0 + \frac{1}{2}Kx_{max}^2$

$\boxed{x_{max} = u\sqrt{\frac{m}{K}}}$

$W_{ext} = 0 = W_{NC}$

Q

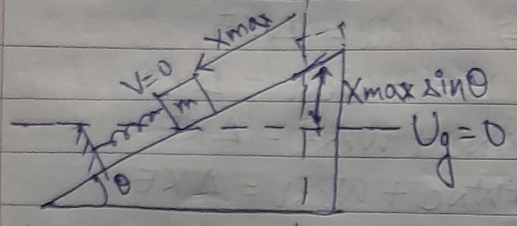


① Find max^m compression in spring.

$= K_i + U_i = K_f + U_f$

$= 0 + 0 + mg x_{max} \sin\theta = 0 + \frac{1}{2} k (x_{max})^2 + 0$

$$\boxed{x_{max} = \frac{2mg\sin\theta}{K}}$$



$U_f = 0$

② Find max^m speed of block.

Max speed is when, $mg\sin\theta = Kx$

$$\boxed{x = \frac{mg\sin\theta}{K}}$$

$\Rightarrow K_i + U_i = K_f + U_f \Rightarrow 0 + 0 + mgx\sin\theta = \frac{1}{2} m v_{max}^2 + \frac{1}{2} k x^2 + 0$

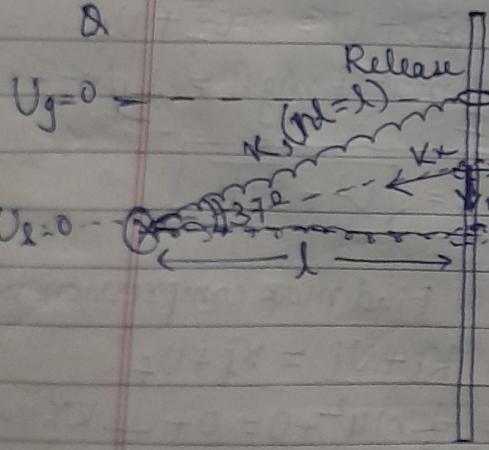
$= mg\left(\frac{mg\sin\theta}{K}\right)\sin\theta = \frac{1}{2} m v_{max}^2 + \frac{1}{2} K \left(\frac{mg\sin\theta}{K}\right)^2$

$= \frac{m^2 g^2 \sin^2\theta}{K} = \frac{1}{2} m v_{max}^2 + \frac{1}{2} \frac{m^2 g^2 \sin^2\theta}{K}$

$= \frac{1}{2} \frac{m^2 g^2 \sin^2\theta}{K} = \frac{1}{2} m v_{max}^2$

$= \boxed{v_{max} = g\sin\theta \sqrt{\frac{m}{K}}}$

Q



Find speed of bead when spring becomes horizontal.

$\Rightarrow K_i + U_i = K_f + U_f$

$= 0 + 0 + \frac{1}{2} K \left(\frac{l}{\cos 37^\circ} - l\right)^2 = \frac{1}{2} m v^2 + mg\left(l + l\tan 37^\circ\right) + 0$

$= \frac{1}{2} K \left(\frac{l^2}{16}\right) = \frac{1}{2} m v^2 + mg\left(\frac{3l}{4}\right)$

$= \frac{Kl^2}{32} = \frac{1}{2} m \left(v^2 + \frac{3gl}{2}\right)$

$$\boxed{v = \sqrt{\frac{3gl}{2} + \frac{Kl^2}{16m}}}$$

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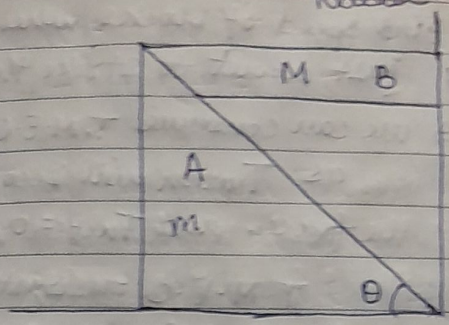
Q



Find speed of block A when block B descends by distance 'h'.

Here work done by N and N' is zero as displacement is in ⊥ dirⁿ. But work done by N'' is not zero and hence it acts as an external force and so T.M.E.C is not valid for A or B individually.

Now lets check for system (A+B)

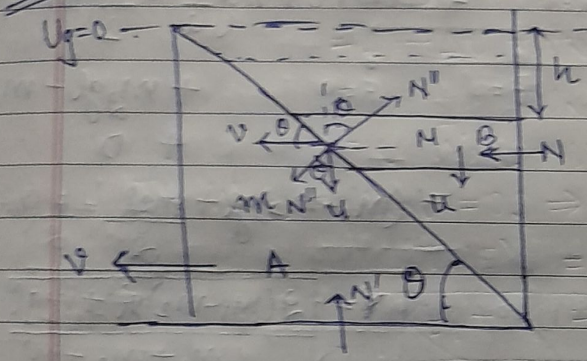
Power of N'' on A = $N''v\cos(90-\theta)$
$= N''v\sin\theta$

Power of N'' on B = $N''u\cos(180-\theta)$
$= -N''v\sin\theta$

Power of N'' on (A+B) = 0

⇒ Work done by N'' on (A+B) = 0

∴ TMEC can be applied on (A+B)



Using, wedge constraint

$v\sin\theta = u\cos\theta$

$\boxed{u = v\tan\theta}$

Using T.M.E.C on system:

$K_i + U_i = K_f + U_f$

$0 + 0 + 0 = \frac{1}{2}mv^2 + \frac{1}{2}Mv^2\tan^2\theta - Mgh$

$Mgh = \frac{1}{2}mv^2 + \frac{1}{2}Mv^2\tan^2\theta$

$$\boxed{V = \sqrt{\frac{2MgH}{(m+M\tan^2\theta)}}}$$

Alternate

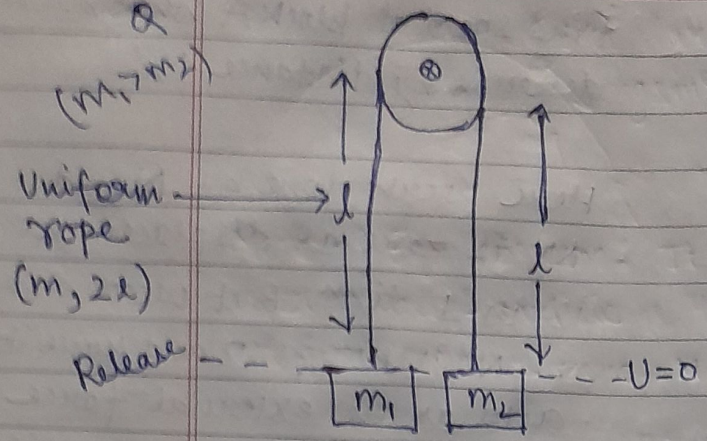
Loss in P.E = Gain in K.E

$MgH = \frac{1}{2}mv^2 + \frac{1}{2}Mv^2\tan^2\theta$

$$\boxed{V = \sqrt{\frac{2MgH}{m+M\tan^2\theta}}}$$

smooth pulley

Q $(m_1 > m_2)$



Find speed of blocks when m_2 is just about to strike the pulley.

⇒ We can conserve T.M.E.C on system as Tension will become internal force and net = 0.

T.M.E.C on system rope + blocks

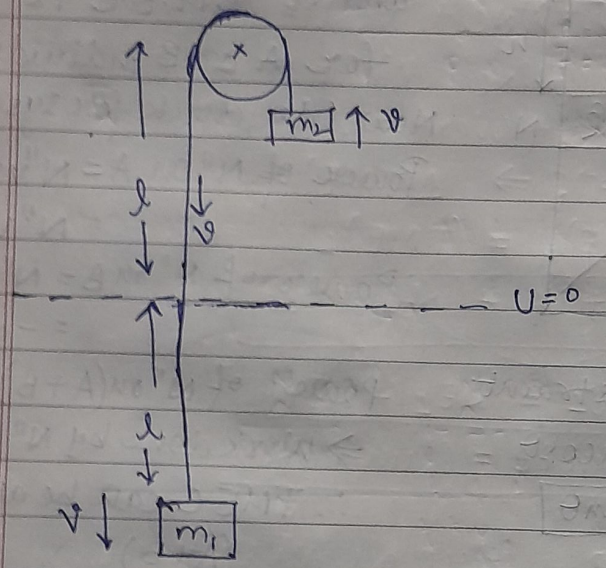
$K_i + U_i = K_f + U_f$

$= 0 + 0 + 0 + mg\frac{l}{2} = \frac{1}{2}(m + m_1 + m_2)v^2$

$+ m_2 gl - m_1 gl$

$+ 0$

$= (m_1 - m_2)gl + \frac{mgl}{2} = \frac{1}{2}(m + m_1 + m_2)v^2$



$$V = \sqrt{\frac{2(m_1 - m_2)gl + mgl}{(m_1 + m_2 + m)}}$$

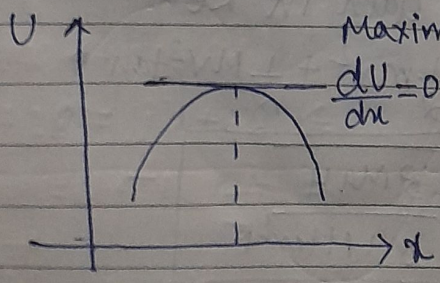
Equilibrium (in presence of conservative forces):

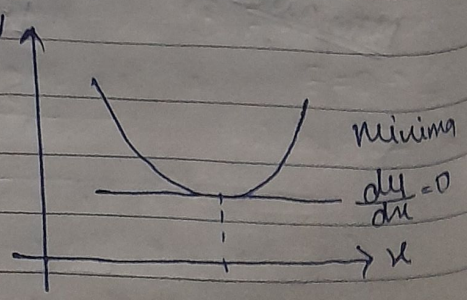
At equilibrium, $\vec{F} = 0$ $\quad \vec{a} = 0$ $\quad \vec{v} = \text{const.}$

$\vec{F} = 0 \Rightarrow -\frac{dU}{dx} = 0 \Rightarrow \frac{dU}{dx} = 0$

At equilibrium ∴ $\boxed{\frac{dU}{dx} = 0}$ ⇒ Either potential energy max^m or min^m ie. U-x curve's maxima or minima

Ex





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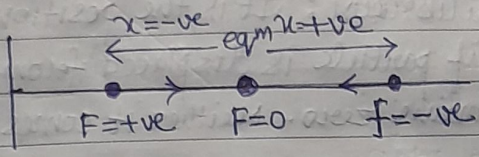
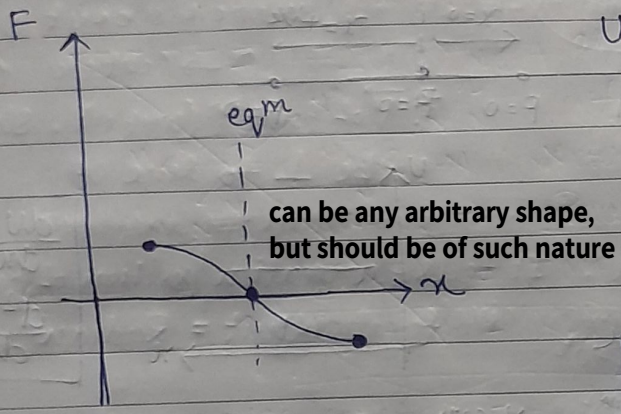
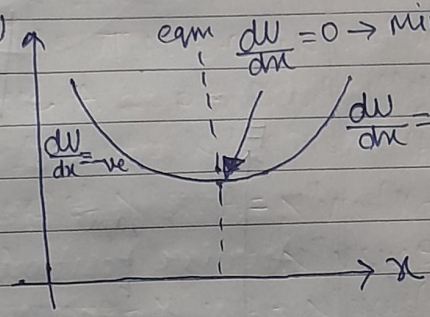
(a) Stable equilibrium : It is the position at which force is zero and particle is displaced from it then net force will act opposite to displacement and this net force tries to restore the equilibrium position.



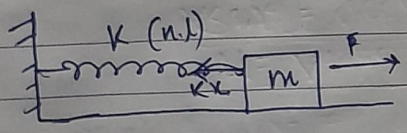


can be any arbitrary shape, but should be of such nature

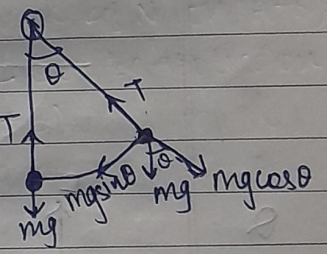


$U_{minima} \Rightarrow \frac{dU}{dx} = 0, \frac{d^2U}{dx^2} > 0$

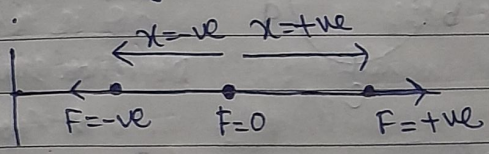
Ex

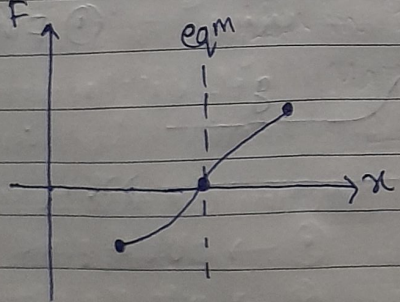


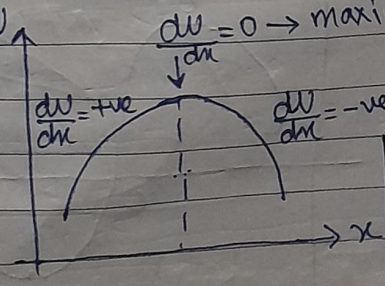
Ex



(b) Unstable Equilibrium : It is a position at which force is zero and if particle is displaced slightly from this position then net force will act along displacement and the particle will run away.



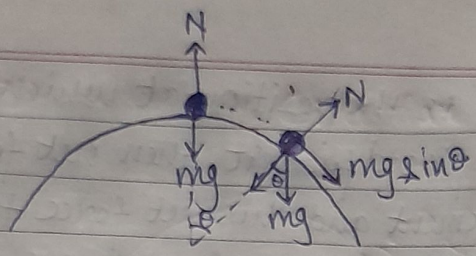




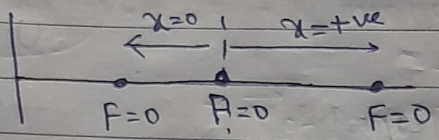
$U_{max} \Rightarrow \frac{dU}{dx} = 0$

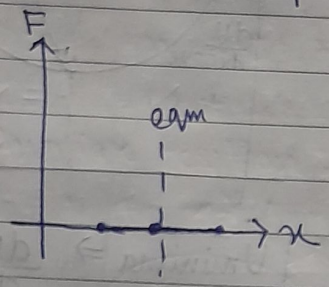
$\frac{d^2U}{dx^2} < 0$

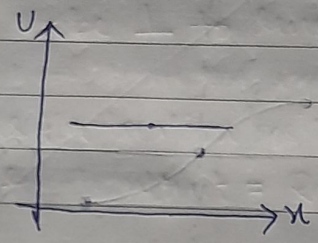
Ex



Neutral Equilibrium: It is the position at which net force is zero and if particle is displaced from this position then also net force remains zero. Neither it comes back nor it runs away.



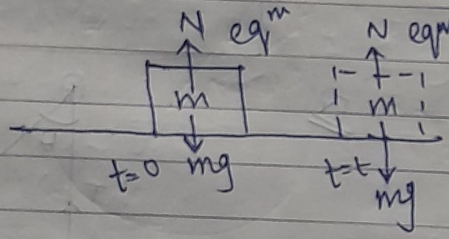




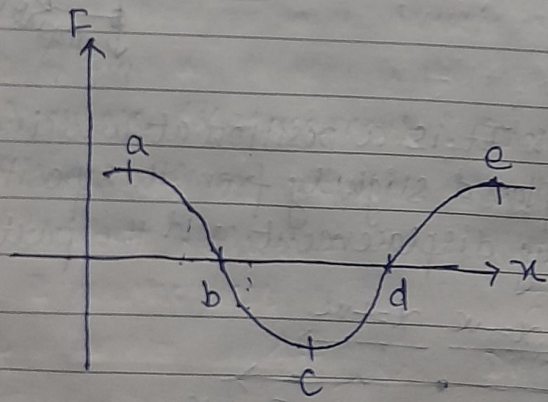
$$\frac{dU}{dx} = 0$$

$$\frac{d^2U}{dx^2} = 0$$

Ex



Q



① Identify equilibrium positions.

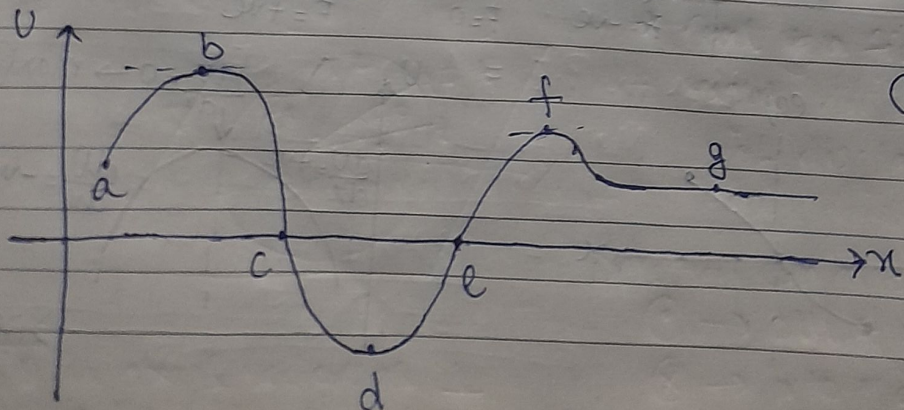
Ans – b and d as F=0

② Also find type of eqm.

b → stable

d → unstable

Q



① Find eqm positions.

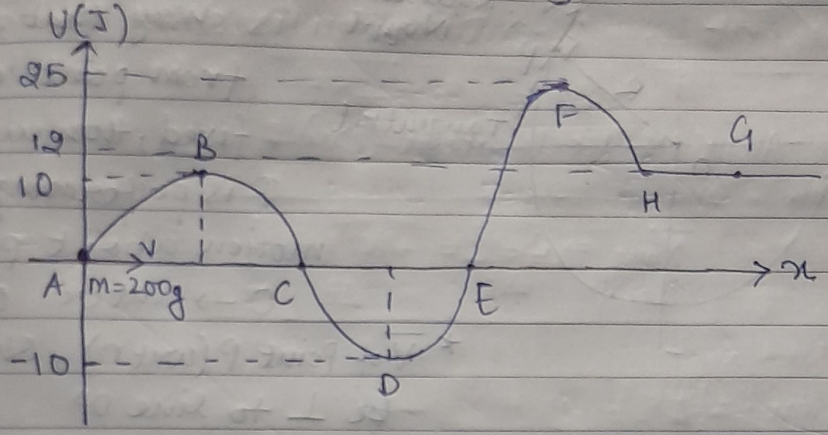
$F=0 \Rightarrow \frac{dU}{dx}=0$

∴ b, d, f and g

② Also find type of equilibrium.

b → unstable
d → stable
f → unstable
g → neutral

Q



① Find min^m v given at origin, so that particle can reach point G.

From A→B : $\frac{dU}{dx}$ = +ve ⇒ F = -ve, so v decreases
From B→D : $\frac{dU}{dx}$ = -ve ⇒ F = +ve, so v increases

From D→F : $\frac{dU}{dx}$ = +ve ⇒ F = -ve, so v decreases
From F→H : $\frac{dU}{dx}$ = -ve ⇒ F = +ve, so v increases
From H→G : $\frac{dU}{dx}$ = 0 ⇒ F = 0, so v is constant.

U_{max} is at F so if we reach F, then v will obviously ↑ and then become const. So min^m v required at F is 0^+

So, apply T.M.E.C b/w A and F:

$$\frac{1}{2} \times 0.2 \times v^2 + 0 = 0^+ + 25$$

$$v^2 = \frac{25}{0.1} = 250$$

$$\boxed{v = 5\sqrt{10}\ m/s}$$

② In this case, find speed of particle when it reaches point G.

T.M.E.C b/w A and G: $\frac{1}{2} \times 0.2 \times 250 + 0 = \frac{1}{2} \times 0.2 \times v^2 + 12$

$$\boxed{v = \sqrt{130}\ m/s}$$

③ Find max^m speed in the given motion.

Max speed will be at D as U is min^m ⇒ KE is max^m.

T.M.E.C b/w A and D ⇒ $\frac{1}{2} \times 0.2 \times 250 + 0 = \frac{1}{2} \times 0.2 \times v_{max}^2 - 10$

$$\boxed{v_{max} = \sqrt{350}\ m/s}$$